
The Collapse Model: Spectral Geometry and Auditable Data-Pool Screening

Anonymous Author(s)
Affiliation
Address
email

Abstract

Given M frozen-representation data pools, which candidates can be short-listed without labels before expensive evaluation? We separate exact spectral claims from empirical screening. For fixed, label-independent preprocessing and declared source weights, the merged spectrum is exactly determined by a sum of positive-semidefinite Gram operators. Under explicit block-compatible singular directions this sum has an exact local 2×2 decomposition; a constructive counterexample shows that marginal spectra and principal angles—and hence the earlier four-scalar summary—do not determine merged effective rank in general. Our primary Stage-1 method is a composable rank- L PSD Gram sketch. Transmitted tail statistics give deterministic Fannes–Audenaert intervals for log-effective-rank error and a sufficient certificate for one Full-Gram greedy step. Controlled sweeps across four encoders show that all 2,880 selected-prefix intervals cover the exact score, but none of 7,200 greedy steps is certified for $L \leq 50$, and DPP is at least as competitive. In 28 held-out encoder-target units, even validation-oracle source-supervised adapters trail frozen identity by 0.067 (target-cluster 95% CI $[-0.098, -0.044]$); small shortlist regret is therefore not downstream benefit. We make no global-optimality, privacy, practical-compression, or downstream-benefit claim. Code is available at <https://anonymous.4open.science/r/collapsemodel-1DB6>.

1 Introduction

Modern pretrained models increasingly rely on combining features from many candidate datasets—multi-domain transfer learning, data curation for self-supervised pretraining [Gadre et al., 2023], and continual learning all face the question: given M candidate datasets, which $k \ll M$ subset preserves the diversity of the full pool when their features are concatenated? We call this the label-free data-pool compression problem: unlike classical coreset selection [Mirzasoileiman et al., 2020] and recent sample-level data curation methods (DSIR [Xie et al., 2023], DsDm [Engstrom et al., 2024], D² Pruning [Maharana et al., 2023]), we seek compression at the dataset level (not the sample level), without access to labels or downstream tasks. The criterion is representational diversity, formalized as the effective rank $r_{\text{eff}}(H) = \exp(-\sum_i p_i \log p_i)$ of the merged features [Roy and Vetterli, 2007, Garrido et al., 2023]. No prior work characterizes how dataset-level spectra and relative subspace geometry jointly affect r_{eff} under feature concatenation, forcing practitioners to rely on exhaustive search or heuristics that scale poorly with M .

Prior work addresses related but distinct problems. Spectral diversity metrics such as RankMe [Garrido et al., 2023], VNE [Kim et al., 2023], the Vendi Score [Friedman and

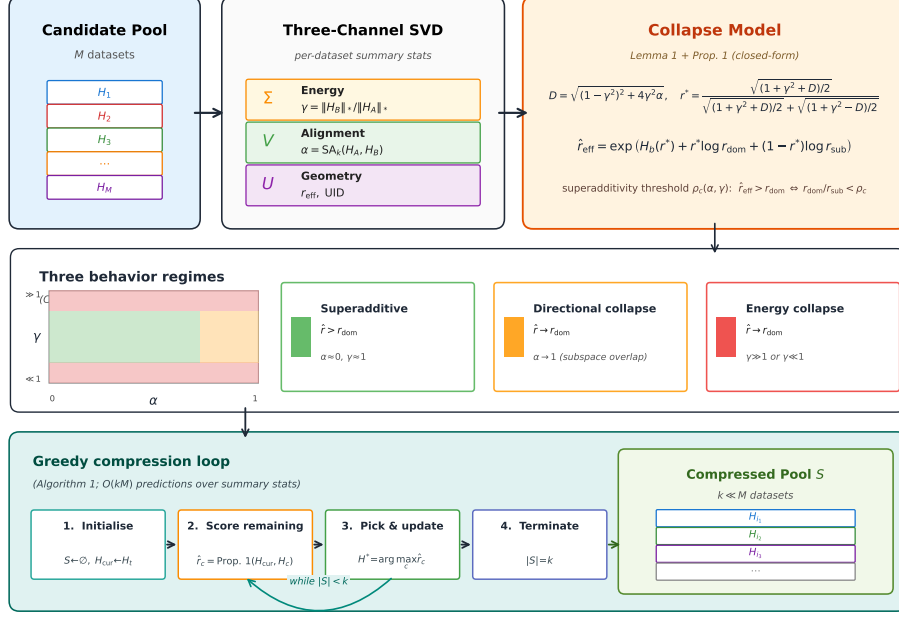


Figure 1: Spectral mechanism and selection abstraction. Top/middle: the paired-direction special case and Collapse-4S diagnostic; these panels explain local behaviour but are not a general guarantee. Bottom: the common greedy loop. Our primary selector scores this loop by composing rank- L PSD factors; Collapse-4S is retained as an ablation.

Dieng, 2023], and its generalized form [Pasarkar and Dieng, 2024] quantify the diversity of a single set via entropy of a singular-value or similarity-matrix spectrum; they do not predict how two sets’ diversity combines under concatenation. Label-free coreset selection (ELFS [Shen et al., 2025], ZCore [Griffin et al., 2024], DEFT-UCS [Das and Khetan, 2024]) selects a subset of samples using geometry or proxy dynamics; it is complementary but orthogonal to our dataset-level formulation. Pool-level curation (DataComp [Gadre et al., 2023], DataComp-LM [Li et al., 2024], DsDm [Engstrom et al., 2024], Quad [Zhang et al., 2025]) uses filtering or model-aware heuristics without a closed-form diversity criterion. Representation similarity measures (SVCCA [Raghu et al., 2017], PWCCA [Morcos et al., 2018], CKA [Kornblith et al., 2019], PARC [Bolya et al., 2021]) compare two representations but do not predict merged rank. Transferability estimators (LogME [Zhang et al., 2021], H-score [Bao et al., 2019], TransRate [Huang et al., 2022], LEEP [Li et al., 2021], reassessed by Ibrahim et al. [2022]) and PMI-based dataset valuation [Zhang et al., 2024] predict downstream accuracy but either require labels or do not model merging. Task2Vec [Achille et al., 2019] avoids labels but requires probe fine-tuning. Model-merging methods (Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], DARE [Yu et al., 2024]) combine parameters, not features. Intrinsic-dimensionality work [Pope et al., 2021, Ansuini et al., 2019] characterizes individual matrices but not merged ones. To our knowledge, no existing work combines an exact local spectral analysis of $r_{\text{eff}}([H_A; H_B])$ with a summary-statistic predictor for dataset-level selection.

We start from the exact identity $G_{A \cup B} = G_A + G_B$ and keep two approximations conceptually separate. The earlier four-summary predictor, which we call Collapse-4S, compresses a paired-direction model into $(r_A, r_B, \gamma, \alpha)$. It is useful for mechanism diagnostics and as an empirical baseline, but Proposition 5 proves that these summaries are not sufficient in general. The primary computational method, Collapse-Gram- L , instead transmits a rank- L PSD factor and certified tail statistics for every source. These summaries compose by addition, retain spectral pairing within the retained subspace, and provide a deterministic error interval (Theorem 1).

Why summary-based screening? Exhaustive search requires $\binom{M}{k}$ merged decompositions and access to all feature rows. Collapse-Gram- L uses $O(kM)$ candidate evaluations and communicates $O(Ld)$ values plus three tail statistics per source. It does not require raw sample features at the coordinator, but this compression is not a differential-privacy guarantee.

Contributions. (1) Exact algebra and an information limit: an unconditional Gram representation, an exact block-compatible decomposition, and an explicit counterexample to low-dimensional identifiability (Propositions 1–5). (2) Certified compression: a composable rank- L PSD sketch with a tail-dependent log-effective-rank interval and a sufficient certificate for matching one step of Full-Gram greedy (Theorem 1, Corollary 6). (3) An audited two-stage protocol: label-blind Stage-1 sampling, declared equal-source weighting, target-isolated Stage-2 evaluation, matched overlap diagnostics, source-independent controls, and provenance-checked resume semantics (Sec. 5).

Empirical scope. The existing pairwise experiments evaluate Collapse-4S, not the new rank- L guarantee. They show strong ordering correlation on the evaluated pairs, but only a 0.003 Spearman improvement over $r_A + r_B$ and encoder-dependent calibration. Historical pool experiments likewise show that spectral methods often select similar clean-pool subsets. We retain these results as bounded evidence and report corrected rank- L , matched overlap, and task-aware comparisons separately rather than treating the old tables as validation of Theorem 1.

2 Spectral Background

For a feature matrix $H \in \mathbb{R}^{N \times d}$ with compact SVD $H = U\Sigma V^\top$, let $\Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_r > 0)$.

Σ channel — spectral energy. The effective rank summarizes spectral distribution:

$$r_{\text{eff}}(H) = \exp\left(-\sum_{i=1}^r p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (1)$$

where p_i is the singular-value share (not squared), so $r_{\text{eff}} \in [1, r]$. The nuclear norm $\|H\|_* = \sum_i \sigma_i$ is the normalizing mass.

V channel — directional structure. Right singular vectors V encode which directions are activated.

Definition 1 (Subspace Alignment). Let $\hat{V}_A^k, \hat{V}_B^k \in \mathbb{R}^{d \times k}$ be the top- k right singular vectors of H_A, H_B . The Subspace Alignment at level k is:

$$\text{SA}_k(H_A, H_B) = \frac{1}{k} \left\| \hat{V}_A^{k\top} \hat{V}_B^k \right\|_F^2 = \frac{1}{k} \sum_{i=1}^k \cos^2 \theta_i, \quad (2)$$

where $\theta_1 \leq \dots \leq \theta_k$ are the principal angles.

$\text{SA}_k \in [0, 1]$ summarizes top-subspace overlap, but discards the principal-vector basis and its coupling to singular values. This loss of information is central to Proposition 5.

3 The Collapse Model

3.1 Setup and Notation

Let $H_A \in \mathbb{R}^{N_A \times d}$ and $H_B \in \mathbb{R}^{N_B \times d}$ be L2-row-normalized feature matrices with effective ranks $r_A = r_{\text{eff}}(H_A)$, $r_B = r_{\text{eff}}(H_B)$, nuclear norms $\|H_A\|_*$, $\|H_B\|_*$, and subspace alignment $\alpha = \text{SA}_k(H_A, H_B)$. Define:

$$\gamma = \frac{\|H_B\|_*}{\|H_A\|_*} \in (0, \infty), \quad r_{\text{dom}} = \begin{cases} r_A & \gamma \leq 1 \\ r_B & \gamma > 1 \end{cases}, \quad r_{\text{sub}} = \begin{cases} r_B & \gamma \leq 1 \\ r_A & \gamma > 1 \end{cases}. \quad (3)$$

γ is the energy ratio of the two matrices; r_{dom} is the effective rank of the higher-energy matrix.

All source preprocessing is fixed before subset selection. For source j with N_j rows, we declare a total source weight $\beta_j > 0$ and use $X_j = \sqrt{\beta_j/N_j} H_j$. Our unequal-size default is $\beta_j = 1$ (equal total mass per source); $\beta_j = N_j$ is the declared equal-per-sample alternative. Although effective rank is invariant to a common scale, it is not invariant to changing relative source weights.

3.2 Main Results

Our analysis separates three levels that must not be conflated: an unconditional Gram identity, an exact paired-direction model under explicit decoupling assumptions, and a low-dimensional summary predictor for general spectra.

Proposition 1 (Exact Gram Representation). Let $H_A = U_A \Sigma_A V_A^\top$ and $H_B = U_B \Sigma_B V_B^\top$ be compact SVDs, and complete V_A to an orthogonal basis $Q_A = [V_A \ V_{A,\perp}]$. With $C = Q_A^\top V_B$ and $\tilde{\Sigma}_A^2$ denoting Σ_A^2 padded by zeros to dimension d , the merged Gram operator satisfies

$$Q_A^\top (H_A^\top H_A + H_B^\top H_B) Q_A = \tilde{\Sigma}_A^2 + C \Sigma_B^2 C^\top. \quad (4)$$

Consequently, the merged singular values are the square roots of the eigenvalues of the right-hand side.

Equation (4) is unconditional. In general C is dense: a singular direction of one matrix can couple to several directions of the other. Individual spectra and an average principal-angle statistic therefore do not determine the merged spectrum.

Lemma 2 (Exact: 2×2 Gram Block Eigenstructure). Assume paired, decoupled directions: for each i , $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, where the two-dimensional planes $\text{span}\{V_{A,i}, V_{\perp,i}\}$ are mutually orthogonal. Define the mode-wise energy ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$ and $\alpha_i = \cos^2 \theta_i$. The i -th Gram block has eigenvalues

$$\lambda_{\pm,i} = \frac{\sigma_{A,i}^2}{2} (A_i \pm D_i), \quad A_i = 1 + \gamma_i^2, \quad D_i = \sqrt{(1 - \gamma_i^2)^2 + 4\gamma_i^2 \alpha_i}. \quad (5)$$

The corresponding singular values are $\sqrt{\lambda_{+,i}}$ and $\sqrt{\lambda_{-,i}}$. This result is exact under the stated pairing and decoupling conditions (Appendix B).

The global ratio $\gamma = \|H_B\|_* / \|H_A\|_*$ does not imply $\gamma_i = \gamma$. Replacing the collection $\{\gamma_i, \alpha_i\}$ by global summaries (γ, α) is an approximation step.

Proposition 3 (Exact Proportional-Spectrum Model). Under Lemma 2, further assume $\sigma_{B,i} = \gamma \sigma_{A,i}$ and $\alpha_i = \alpha$ for every i . Let $A = 1 + \gamma^2$, $D = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2 \alpha}$, and

$$q = \frac{\sqrt{(A+D)/2}}{\sqrt{(A+D)/2} + \sqrt{(A-D)/2}}. \quad (6)$$

Then the merged spectrum consists of two scaled copies of the spectrum of H_A , and

$$r_{\text{merged}} = r_A \exp(H_b(q)), \quad r_A = r_B. \quad (7)$$

Definition 2 (Collapse Summary Predictor). For general spectra, we compress the mode-wise quantities to the global nuclear-norm ratio γ and mean top- k alignment α , compute q from Eq. (6), and define

$$\hat{r}_{\text{merged}} = \exp(H_b(q) + q \log r_{\text{dom}} + (1 - q) \log r_{\text{sub}}), \quad (8)$$

where $H_b(p) = -p \log p - (1 - p) \log(1 - p)$.

Equation (8) is a theory-motivated low-dimensional approximation, not an identity or a universally bounded estimator. It interpolates the input spectral entropies and adds the binary entropy of the idealized two-branch split. Its accuracy on general spectra is an empirical question evaluated in Sec. 4.

Interpretation. Equation (8) is a weighted log-geometric mean of the individual ranks plus a balance bonus $H_b(q)$ (maximal when $q=0.5$, vanishing as $q \rightarrow 0, 1$).

Behavior regimes. Superadditive ($\alpha \approx 0, \gamma \approx 1$): $q \approx 0.5$, $\hat{r}_{\text{merged}} \approx 2\sqrt{r_A r_B} \geq r_{\text{dom}}$. Directional collapse ($\alpha \rightarrow 1$): $q \rightarrow 1$, $H_b(q) \rightarrow 0$, $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. Energy collapse ($\gamma \gg 1$ or $\gamma \ll 1$): $q \rightarrow 1$, $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. When $\alpha \rightarrow 1$ and $\gamma \gg 1$, the merged rank can fall below r_{sub} —a net loss in diversity.

3.3 Superadditivity Threshold

Corollary 4 (Summary-Model Superadditivity Diagnostic). The summary predictor in Eq. (8) predicts $\hat{r}_{\text{merged}} > r_{\text{dom}}$ for $q < 1$ if and only if

$$\frac{r_{\text{dom}}}{r_{\text{sub}}} < \rho_c(\alpha, \gamma) \equiv \exp\left(\frac{H_b(q)}{1-q}\right), \quad (9)$$

where q is from Eq. (6). The threshold ρ_c is increasing in α : $\rho_c \geq 4.0$ for all $\alpha \in [0, 0.9]$ at $\gamma = 1$. As $q \rightarrow 1^-$, $\rho_c \rightarrow \infty$ while the predicted gain tends to zero. At the degenerate point $q = 1$ the division in Eq. (9) is undefined and Eq. (8) gives $\hat{r}_{\text{merged}} = r_{\text{dom}}$ exactly.

This is an exact decision boundary for the summary model, and for the proportional-spectrum model of Proposition 3; on general data it is a diagnostic rather than a guarantee.

Physical meaning. ρ_c is a decision threshold internal to the summary model. Near $q = 1$, it can classify an arbitrarily small positive entropy increment as superadditive; the magnitude $\hat{r}_{\text{merged}} - r_{\text{dom}}$ must therefore be reported alongside the binary diagnosis. Exact directional overlap gives $q = 1$ and no predicted gain. The proofs and model scope are in Appendices B, L.

3.4 Marginal Spectra and Principal Angles Are Not Sufficient

Proposition 5 (Non-identifiability from Marginal Geometry). Even the two marginal singular spectra together with the complete set of principal angles between the right singular subspaces do not uniquely determine $r_{\text{eff}}([H_A; H_B])$. Consequently, neither does the tuple $(r_A, r_B, \gamma, \alpha)$.

Counterexample. Let e_1, e_2, e_3 be standard basis vectors and define the row-normalized matrices

$$H_A = [e_1, e_1, e_1, e_1, e_2]^\top, \quad H_B^{(1)} = [e_1, e_1, e_1, e_1, e_3]^\top, \quad H_B^{(2)} = [e_1, e_3, e_3, e_3, e_3]^\top. \quad (10)$$

All three have nonzero singular values $(2, 1)$. Both right-subspace pairs have principal angles $\{0, \pi/2\}$, yet their merged Gram spectra are $(8, 1, 1)$ and $(5, 4, 1)$, respectively. The corresponding effective ranks are 2.626012 and 2.849536. Thus even complete marginal spectra and principal angles omit how spectral mass is assigned to coupled directions; the four scalar summaries discard still more information. $\square$

3.5 Composable Rank- L Gram Sketch

The counterexample motivates approximating the PSD operator itself rather than compressing it to averaged angles. For the compact SVD of each preweighted source, define

$$B_j = \Sigma_{j,1:L} V_{j,1:L}^\top, \quad \tilde{G}_j = B_j^\top B_j, \quad E_j = G_j - \tilde{G}_j \succeq 0. \quad (11)$$

Attach the tail statistics $\tau_j = \text{tr} \sqrt{E_j}$, $\omega_j = \text{tr}(E_j)$, and $u_j = \text{rank}(E_j)$; certified upper bounds may replace their exact values. For a subset S , vertically stack the factors into B_S . Then $\tilde{G}_S = B_S^\top B_S$, so its nonzero spectrum is obtained from the small core $B_S B_S^\top$ without reconstructing sample features.

For a PSD matrix G , write $\mathcal{R}(G) = \exp \mathcal{H}(\sqrt{\lambda(G)}/\text{tr} \sqrt{G})$; hence $\mathcal{R}(H^\top H) = r_{\text{eff}}(H)$. Define

$$T_S = \min \left\{ \sum_{j \in S} \tau_j, \sqrt{\min(d, \sum_{j \in S} u_j) \sum_{j \in S} \omega_j} \right\}, \quad \varepsilon_S = \frac{T_S}{\text{tr} \sqrt{\tilde{G}_S} + T_S}. \quad (12)$$

Theorem 1 (Rank- L Log-Effective-Rank Bound). Assume $\tilde{G}_S \neq 0$, and let $D_S = \min(d, \sum_{j \in S} \text{rank}(G_j))$. For $D \geq 2$, set

$$\Phi_D(\varepsilon) = \begin{cases} h_b(\varepsilon) + \varepsilon \log(D-1), & 0 \leq \varepsilon \leq 1 - 1/D, \\ \log D, & 1 - 1/D < \varepsilon \leq 1. \end{cases} \quad (13)$$

Table 1: E1a: Synthetic grid results ($N = 280$). Naive baseline fails at large γ .

Method	R^2	Max error	Superadd. (%)	$\alpha < 1$ superadd.
Naive $\sqrt{1 + \alpha\gamma^2}$	0.30	28.73	—	—
Collapse-4S	0.9999	0.27	87.5%	100% (245/245)

182 Then, without a block-compatibility assumption,

$$\left| \log \mathcal{R}(G_S) - \log \mathcal{R}(\tilde{G}_S) \right| \leq \Phi_{D_S}(\varepsilon_S). \quad (14)$$

183 For $D_S = 1$ the error is zero.

184 Corollary 6 (One-Step Full-Gram Certificate). At a greedy step, let $\hat{F}_j = \log \mathcal{R}(\tilde{G}_{S \cup \{j\}})$ and
 185 $e_j = \Phi_{D_{S \cup \{j\}}}(\varepsilon_{S \cup \{j\}})$. If a candidate j^* satisfies

$$\hat{F}_{j^*} - e_{j^*} > \max_{j \neq j^*} (\hat{F}_j + e_j), \quad (15)$$

186 then j^* uniquely maximizes the exact Full-Gram score in that round.

187 The interval accumulates tails from the selected set and the candidate; using only the
 188 candidate tail is invalid. Failure to separate intervals is a diagnostic to increase L or fall
 189 back to Full-Gram scoring. The certificate is local to one greedy step: effective rank is
 190 generally nonmonotone and nonsubmodular, so it gives no global subset-optimality claim
 191 (Appendix C).

192 Legacy Collapse-4S calibration. On the evaluated real-data pairs Eq. (8) systematically
 193 over-predicts. A fitted multiplicative constant reduces average scale error; the reported
 194 relation $c=0.946-0.175 \log(N/r_{\text{eff}})$ is an empirical fit over the evaluated encoder families,
 195 not a universal law (Appendix L).

196 4 Evaluation of the Legacy Four-Summary Predictor

197 This section reports the historical evaluation of Collapse-4S. These experiments test Eq. (8),
 198 not the rank- L approximation or its certificate. We organize the evidence as: (i) idealized-
 199 model agreement—the predictor agrees with data generated under its summary assump-
 200 tions (§4.1, synthetic grid); (ii) real-data association—the formula remains correlated with
 201 approximation across 5 encoder families, 14 vision datasets, and 19 text corpora (§4.2,
 202 427 pairs); (iii) behavioral regime agreement—empirical superadditivity patterns match the
 203 three regimes predicted by Corollary 4 (§4.2, Table 2). Together these locate useful and fail-
 204 ing regimes of the heuristic. Additional controls (E1a-Ind 120 non-principal-angle pairs, E1b
 205 30 real pairs, E1c 57 pairs, E_downstream cross-architecture, E_downstream r_{eff} /LP-accuracy)
 206 all agree with the same picture and are deferred to Appendix M. Feature matrices are L2-
 207 row-normalized and subsampled to N_{sub} samples; effective rank uses Eq. (1); bootstrap CIs
 208 use $B = 10,000$ resamples.

209 4.1 Step 1 — Idealized-Model Agreement: Synthetic Controlled Grid (E1a, 280 pairs)

210 As a within-model sanity check, we evaluate Collapse-4S on 280 controlled synthetic
 211 pairs generated under the proportional-spectrum, paired-direction model with $\alpha \in$
 212 $\{0, 0.05, \dots, 1.0\}$ and $\gamma \in \{0.1, 0.33, \dots, 10.0\}$ ($d=512$, $N=1000$, 5 repeats per setting).

213 Table 1 shows $R^2=0.9999$ (vs. 0.30 for the naive baseline). All 245 configurations with
 214 $\alpha < 1$ achieve 100% superadditivity; all 35 non-superadditive cases concentrate exclusively at
 215 $\alpha=1$ (identical subspaces), as expected from the generator and the boundary of Corollary 4.
 216 Visualization of the (α, γ) heatmap and superadditivity region is in Appendix M.

217 4.2 Steps 2–3 — Real-Data Generalization and Regime Agreement (E1d, 427 Pairs)

218 We next measure association beyond the synthetic derivation and test whether the predicted
 219 three-regime behavior is observable. The evaluation spans 5 encoder families (ResNet-50,

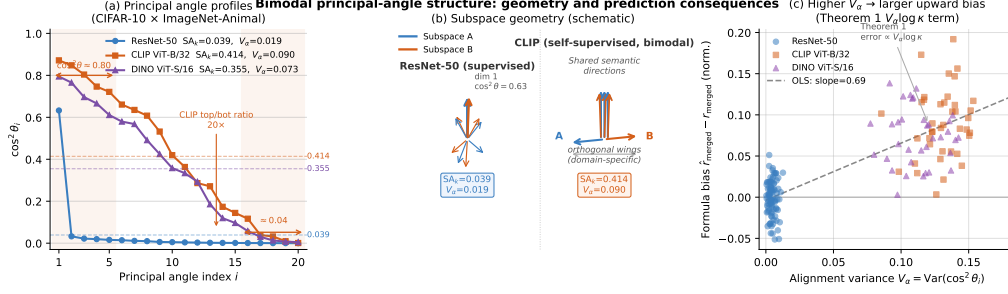


Figure 2: Bimodal principal-angle structure (CIFAR-10 $\times$ ImageNet-Animal, $k=20$). (a) Per-dimension $\cos^2 \theta_i$: ResNet-50 ($SA_k=0.039, V_\alpha=0.019$) decays monotonically from one dominant direction; CLIP ($SA_k=0.414, V_\alpha=0.090$) and DINO ($SA_k=0.355, V_\alpha=0.073$) form a two-cluster step (top-5 ratios $20\times/31\times$ over bot-5), violating the uniform- α assumption. (b) Geometric interpretation: one shared spine plus orthogonal domain-specific wings. (c) Alignment variance V_α predicts upward formula bias across 166 pairs (OLS slope 0.69), identifying alignment heterogeneity as an empirical failure indicator.

220 ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100,
 221 STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and
 222 5 ImageNet semantic subsets animal/vehicle/artifact/food/scene), and 19 SBERT text cor-
 223 pora. Total: 427 pairs (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT).

Table 2: E1d: Collapse-4S association study (427 pairs, 5 probes). c is fitted separately per encoder family; calibrated R^2 is therefore descriptive rather than out-of-family generalization.

Probe	n	Superadd. %	Pearson r	c	R^2_{cal}
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	per-probe	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	per-probe	0.952

224 Association (Step 2). Across the 427 evaluated pairs, the formula has overall Pearson
 225 $r=0.970$ (uncalibrated) and $R^2_{\text{cal}}=0.952$ after per-encoder-family calibration (scatter visu-
 226 alization in Appendix M, Fig. 5). The correlation measures ranking on this benchmark;
 227 fitted calibration does not remove the non-identifiability of Proposition 5.

228 Regime agreement (Step 3). (i) ResNet-50 achieves 100% superadditivity across the 91
 229 evaluated pairs (including medical and ImageNet), consistent with the balanced- (γ, α) di-
 230 agnostic of Corollary 4. (ii) Supervised/self-supervised divide emerges in the model’s
 231 directional-overlap regime: CLIP and DINO achieve only 29%–36% superadditivity, with
 232 failures concentrating entirely in cross-domain pairs (all evaluated handwritten $\times$ ImageNet
 233 and natural $\times$ digit pairs fail) while within-domain pairs remain near-perfect. A bimodal
 234 principal-angle distribution and extreme condition numbers $\kappa \in [10^4, 10^6]$ are associated with
 235 these failures (Figure 2). This pattern is consistent with the directional-collapse diagnostic
 236 of Sec. 3.3, but the comparison is not a causal identification.

237 Quantifying the bimodal gap. Per-probe calibrated R^2 : CLIP 0.84 > ResNet 0.76 > DINO
 238 0.55 > ViT 0.49; ranking accuracy is uniformly high ($r \geq 0.90$). Figure 2(a,c) shows that
 239 CLIP/DINO’s two-cluster $\cos^2 \theta_i$ distribution ($V_\alpha > 0.07$) inflates the observed prediction
 240 bias, identifying a concrete departure from the global summary approximation rather than
 241 a black-box failure (full mechanism in Appendix E).

242 (iii) Energy collapse appears in ViT’s 17 non-superadditive pairs, all in $\gamma \notin [0.5, 2]$, matching
 243 the summary model’s energy-imbalance diagnostic. (iv) Cross-modal check: on 171 SBERT

text pairs the ranking correlation drops to $r=0.793$ (98.8% superadditive), so the data do not support a modality-agnostic claim. Absolute calibration, however, requires modality-specific tuning: the vision-only fitted regression ($c=0.946-0.175\log(N/r_{\text{eff}})$, Appendix L) explains $R^2=0.96$ of vision variance but only $R^2=0.08$ of SBERT variance, indicating that text encoders have a qualitatively different spectral bias. A text-specific calibration is left for future work.

Evaluation summary. Steps 1–3 establish within-model agreement, benchmark association, and observable qualitative regimes. They neither validate Collapse-Gram- L nor establish universal exactness: Proposition 5 proves that four summaries cannot determine merged rank in general. Cross-dataset calibration generalization, cross-probe consistency, and r_{eff}/LP -accuracy correspondence ($\rho=0.90$ within ViT family) are in Appendix M.

5 Two-Stage Data-Pool Compression

The objective of Stage 1 is a label-blind shortlist, not a final utility decision. Collapse-Gram- L applies Algorithm 1 to the composable sketches of Sec. 3.5. In the current Stage 2 audit, each candidate adapter is trained with that source’s labels; repeated cross-validation inside target-train fits ridge probes and selects the source. The target-test split is opened once after selection is frozen. This is a source-supervised adaptation protocol with target-validation selection, not a target-trained adapter. The separation is necessary because spectral diversity alone cannot determine arbitrary target utility.

Corrected protocol. Source rows for Stage 1 are sampled without labels, and unequal-size sources use the declared equal-source weighting $X_j = H_j/\sqrt{N_j}$ ($\beta_j = 1$). We compare Collapse-Gram- L against rank-only, DPP, Facility Location, agglomerative medoids, leverage-score selection, random selection, and Full-Gram greedy; an exhaustive merged-rank oracle is included only when the predeclared combination-count limit permits it. All Stage-2 candidates receive identical source-supervised optimization steps and seeds, with frozen-identity and untrained random-adapter controls. Target-train cross-validation is the only target signal used before selection and all adapter checkpoints are hashed. A selection manifest is then written and hashed before a separate held-out command is permitted to open the target-test cache. Manifests bind code, inputs, preprocessing, weights, sampled indices, checkpoints, and hyperparameters by SHA-256; incomplete source/seed groups are never resumed as completed runs. Overlap detection is reported against same-dataset matched negatives, and the 0% overlap condition has no parent ground truth. To estimate exhaustive shortlist regret, the audit trained all candidate adapters offline; its total wall time is therefore not a measured deployment saving.

Audited historical pilot. The pre-correction 21-source pilot used label-stratified source caches, queried target test before freezing selection, and did not evaluate Collapse-Gram- L . Its DPP and rank-only Recall@10 were both 85.71%, with strong near-tie sensitivity; we retain it only as a historical audit baseline and report the corrected results below.

Corrected controlled evidence. An intermediate `corrected_v2` run was invalidated because of a statistic-definition mismatch and incomplete provenance. In the replacement `corrected_v3` sweep, four encoders, ten construction seeds, six overlap levels, and four budgets give 960 configurations per rank. Table 3 reports macro-averages over encoders. Every selected-prefix interval contains the exact score, but all intervals overlap at every evaluated decision. The bound is valid yet operationally vacuous at these ranks. DPP is stronger in this controlled spectral-retention task; this is a recorded unequal-byte comparison, not a communication-matched Pareto result. A separate $L = 20$ exact-reference pilot (four encoders, three seeds, 288 configurations, Random-100) gives rank- L /DPP/Full-Gram gains of 3.65/5.37/5.81% over rank-only. Its intervals cover 288/288 prefixes, but certify 0/720 steps and match 0/288 Full-Gram ordered prefixes; Full-Gram here is a greedy reference, not a global oracle.

Corrected held-out utility evidence. The natural-pool rerun contains 21 sources, 7 targets, 4 encoders, and 5 adapter seeds. At shortlist 10/21, strict oracle recall is 75.00% for rank- L ,

Table 3: Corrected controlled rank sweep. Gains are relative to rank-only in exact final merged effective rank. All 960 intervals per row cover the exact selected-prefix score.

L	Gram gain	DPP gain	Mean error / bound	Certified steps
10	+2.09%	+4.90%	2.558/5.944	0/2,400
20	+3.81%	+5.48%	1.877/5.505	0/2,400
50	+4.94%	+5.68%	0.996/4.455	0/2,400

85.71% for rank-only and DPP, and 89.29% for Full-Gram greedy; none reaches the predeclared 90% threshold. Their held-out regrets within the adapter family are small (0.00022–0.00082), but the family itself fails the source-independent control. Mean held-out accuracy is 0.83654 for frozen identity, 0.77144 for an untrained random adapter, and 0.76951 for the validation-oracle source-supervised adapter. Oracle minus identity is -0.06703 (target-cluster 95% CI $[-0.09771, -0.04400]$); every deterministic selected adapter loses to identity in all 28 encoder-target units. Thus small regret cannot be interpreted as preserved utility or beneficial data use.

Interpretation. The theory controls spectral-score approximation only. The corrected sweep confirms interval coverage but shows that coverage alone need not produce a usable decision certificate. Stage 2 then rejects the current source-supervised adapter family before any practical-compression claim is made. Exact identity, tie-aware recall, regret, and source-independent utility controls are therefore separate quantities.

6 Conclusion

We separated exact spectral structure, bounded approximation, and task-dependent evidence. Gram additivity and the block-compatible local model are exact; the counterexample rules out universal prediction from the legacy summaries. The rank- L sketch has a valid deterministic error interval, but the corrected sweep obtains zero decision certificates for $L \leq 50$ and finds DPP at least as competitive. The held-out audit is more decisive: even the validation-oracle source-supervised adapter is substantially worse than frozen identity. The current contribution is therefore an exact local analysis, an auditable approximation protocol, and a documented negative result—not a validated practical compressor. Equal-byte Pareto evidence and a downstream adapter family that first beats identity are prerequisites for a positive application claim. Code is available at <https://anonymous.4open.science/r/collapsemodel-1DB6>.

References

- A. Achille, M. Lam, R. Tewari, A. Ravichandran, S. Maji, C. C. Fowlkes, S. Soatto, and P. Perona. Task2Vec: Task embedding for meta-learning. In IEEE/CVF International Conference on Computer Vision (ICCV), pages 6430–6439, 2019.
- A. Ansuini, A. Laio, J. H. Macke, and D. Zoccolan. Intrinsic dimension of data representations in deep neural networks. In Advances in Neural Information Processing Systems (NeurIPS), pages 6111–6122, 2019.
- Y. Bao, Y. Li, S.-L. Huang, L. Zhang, L. Zheng, A. Zamir, and L. Guibas. An information-theoretic approach to transferability in task transfer learning. In IEEE International Conference on Image Processing (ICIP), pages 2309–2313, 2019.
- D. Bolya, R. Mittapalli, and J. Hoffman. Scalable diverse model selection for accessible transfer learning. In Advances in Neural Information Processing Systems (NeurIPS), 2021.
- D. Das and V. Khetan. DEFT-UCS: Data efficient fine-tuning for pre-trained language models via unsupervised core-set selection. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.

337 A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. De-
338 hghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is
339 worth 16x16 words: Transformers for image recognition at scale. In International Confer-
340 ence on Learning Representations (ICLR), 2021.

341 L. Engstrom, A. Feldmann, and A. Madry. DsDm: Model-aware dataset selection with
342 datamodels. In International Conference on Machine Learning (ICML), 2024.

343 D. Friedman and A. B. Dieng. The Vendi score: A diversity evaluation metric for machine
344 learning. Transactions on Machine Learning Research (TMLR), 2023.

345 S. Y. Gadre, G. Ilharco, A. Fang, J. Hayase, G. Smyrnis, T. Nguyen, M. Wortsman, et al.
346 DataComp: In search of the next generation of multimodal datasets. In Advances in
347 Neural Information Processing Systems (NeurIPS), 2023.

348 Q. Garrido, R. Balestrieri, L. Najman, and Y. LeCun. Rankme: Assessing the downstream
349 performance of pretrained self-supervised representations by their rank. In ICML, 2023.

350 B. A. Griffin, J. Heikkilä, and J. J. Corso. Zero-shot coreset selection via iterative subspace
351 sampling. arXiv preprint arXiv:2411.15349, 2024.

352 K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In
353 Conference on Computer Vision and Pattern Recognition (CVPR), pages 770–778, 2016.

354 L.-K. Huang, J. Huang, Y. Rong, Q. Yang, and Y. Wei. Frustratingly easy transferability
355 estimation. In International Conference on Machine Learning (ICML), pages 9201–9225,
356 2022.

357 S. Ibrahim, N. Ponomareva, and R. Mazumder. Newer is not always better: Rethinking
358 transferability metrics, their peculiarities, stability and performance. In European Con-
359 ference on Machine Learning (ECML-PKDD), 2022.

360 G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and
361 A. Farhadi. Editing models with task arithmetic. In International Conference on Learning
362 Representations (ICLR), 2023.

363 J. Kim, S. Park, and W. Shin. VNE: An effective method for improving deep representation
364 by manipulating eigenvalue distribution. 2023.

365 S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network repre-
366 sentations revisited. In International Conference on Machine Learning (ICML), pages
367 3519–3529, 2019.

368 J. Li, A. Fang, G. Smyrnis, M. Ivgi, M. Jordan, S. Y. Gadre, H. Bansal, E. Guha, S. Keh,
369 K. Arora, et al. DataComp-LM: In search of the next generation of training sets for lan-
370 guage models. In Advances in Neural Information Processing Systems (NeurIPS) Datasets
371 and Benchmarks Track, 2024.

372 Y. Li, X. Jia, R. Sang, Y. Zhu, B. Green, L. Wang, and B. Gong. Ranking neural checkpoints.
373 In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages
374 2663–2673, 2021.

375 A. Maharana, P. Yadav, and M. Bansal. D2 pruning: Message passing for balancing diversity
376 and difficulty in data pruning. In International Conference on Machine Learning (ICML),
377 2023.

378 B. Mirzsoleiman, J. Bilmes, and J. Leskovec. Coresets for data-efficient training of machine
379 learning models. In International Conference on Machine Learning (ICML), 2020.

380 A. S. Morcos, M. Raghu, and S. Bengio. Insights on representational similarity in neural net-
381 works with canonical correlation. In Advances in Neural Information Processing Systems
382 (NeurIPS), pages 5732–5741, 2018.

- 383 A. P. Pasarkar and A. B. Dieng. Cousins of the Vendi score: A family of similarity-based di-
384 versity metrics for science and machine learning. In International Conference on Artificial
385 Intelligence and Statistics (AISTATS), 2024.
- 386 P. Pope, C. Zhu, A. Abdelkader, M. Goldblum, and T. Goldstein. The intrinsic dimension
387 of images and its impact on learning. In International Conference on Learning Represen-
388 tations (ICLR), 2021.
- 389 M. Raghu, J. Gilmer, J. Yosinski, and J. Sohl-Dickstein. SVCCA: Singular vector canonical
390 correlation analysis for deep learning dynamics and interpretability. In Advances in Neural
391 Information Processing Systems (NIPS), pages 6076–6085, 2017.
- 392 N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-
393 networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Lan-
394 guage Processing (EMNLP), 2019.
- 395 O. Roy and M. Vetterli. The effective rank: A measure of effective dimensionality. pages
396 606–610, 2007.
- 397 H. Shen, P. Murugesan, A. Cidon, O. Zohar, S. Yang, A. Spielberg, A. Krishnamurthy,
398 S. Savarese, and C. Xiong. ELFS: Label-free coreset selection with proxy training dynam-
399 ics. In International Conference on Learning Representations (ICLR), 2025.
- 400 S. M. Xie, S. Santurkar, T. Ma, and P. Liang. Data selection for language models via
401 importance resampling. In Advances in Neural Information Processing Systems (NeurIPS),
402 2023.
- 403 P. Yadav, D. Tam, L. Choshen, C. Raffel, and M. Bansal. TIES-merging: Resolving in-
404 terference when merging models. In Advances in Neural Information Processing Systems
405 (NeurIPS), 2023.
- 406 L. Yu, B. Yu, H. Yu, F. Huang, and Y. Li. Language models are super Mario: Absorbing
407 abilities from homologous models as a free lunch. In International Conference on Machine
408 Learning (ICML), 2024.
- 409 C. Zhang, H. Zhong, K. Zhang, C. Chai, R. Wang, X. Zhuang, T. Bai, J. Qiu, L. Cao,
410 J. Fan, Y. Luo, J. He, and H. Chen. Harnessing diversity for important data selection in
411 pretraining large language models. In International Conference on Learning Representa-
412 tions (ICLR), 2025.
- 413 K. Zhang, K. Shen, J. Han, P. Jochems, X. Zhang, and J. Zhang. LogME: Practical assess-
414 ment of pre-trained models for transfer learning. In International Conference on Machine
415 Learning (ICML), pages 12133–12143, 2021.
- 416 S. Zhang, Y. Kong, and G. Schoenebeck. Proper dataset valuation by pointwise mutual
417 information. In International Conference on Machine Learning (ICML), 2024.

418 A Limitations

419 The four summaries $(r_A, r_B, \gamma, \alpha)$ are not sufficient to determine merged effective rank in
420 general (Proposition 5). The exact 2×2 decomposition requires mode-wise pairing and
421 decoupling; the closed entropy expression additionally requires proportional spectra and
422 uniform alignment. We do not claim a universal error bound for Collapse-4S. The empirical
423 calibration relation (Eq. 25) is fitted on few encoder families and transfers poorly to SBERT.
424 The rank- L theorem requires an exact PSD truncation (or valid tail upper bounds) under
425 fixed, subset-independent preprocessing; data-dependent global centering would break direct
426 composability. Its intervals are too wide to certify any tested choice at $L \leq 50$, and increas-
427 ing L increases communication; the current baselines are not equal-byte matched. Forming a
428 Gram matrix also squares the condition number, so extremely ill-conditioned inputs require
429 direct-SVD cross-checks or explicit numerical slack. Finally, unlabeled geometry cannot
430 determine arbitrary labels. The corrected natural pool remains domain imbalanced, and

the source-supervised bottleneck adapter is a narrow proxy that fails against frozen identity. LoRA and multi-pool training were consequently not run under the preregistered stopping rule. Broader domains and a useful adapter family would be required before claiming practical utility.

B Proofs of the Exact Gram Results

Proposition 1. From the compact SVDs, $H_A^\top H_A = V_A \Sigma_A^2 V_A^\top$ and $H_B^\top H_B = V_B \Sigma_B^2 V_B^\top$. Because the first columns of Q_A are V_A , $Q_A^\top V_A \Sigma_A^2 V_A^\top Q_A = \tilde{\Sigma}_A^2$. Writing $C = Q_A^\top V_B$ gives $Q_A^\top V_B \Sigma_B^2 V_B^\top Q_A = C \Sigma_B^2 C^\top$. Adding proves Eq. (4). Orthogonal similarity preserves eigenvalues, whose square roots are the merged singular values. $\square$

Lemma 2.

Under paired, decoupled directions, the i -th right singular vector of H_B is $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, where $V_{A,i}$ is the i -th right singular vector of H_A and the two-dimensional planes are mutually orthogonal. In the basis $\{V_{A,i}, V_{\perp,i}\}$:

$$H_A^\top H_A \text{ (block)} = \sigma_{A,i}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (16)$$

$$H_B^\top H_B \text{ (block)} = \sigma_{B,i}^2 \begin{pmatrix} \cos^2 \theta_i & \cos \theta_i \sin \theta_i \\ \cos \theta_i \sin \theta_i & \sin^2 \theta_i \end{pmatrix}. \quad (17)$$

Define the mode-wise ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$; no global proportional-spectrum assumption is required. Summing gives

$$G_i = \sigma_{A,i}^2 \begin{pmatrix} 1 + \gamma_i^2 \alpha_i & \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} \\ \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} & \gamma_i^2(1 - \alpha_i) \end{pmatrix}, \quad (18)$$

where $\alpha_i = \cos^2 \theta_i$. Its trace is $\sigma_{A,i}^2(1 + \gamma_i^2)$ and determinant is $\sigma_{A,i}^4 \gamma_i^2(1 - \alpha_i)$, yielding Eq. (5) by the quadratic formula. $\square$

Proposition 3.

With $\gamma_i = \gamma$ and $\alpha_i = \alpha$, define $c_\pm = \sqrt{(A \pm D)/2}$. Lemma 2 gives merged singular values $\{c_+ \sigma_{A,i}\}_i \cup \{c_- \sigma_{A,i}\}_i$. After nuclear-norm normalization their probabilities are qp_i^A and $(1 - q)p_i^A$, where $q = c_+/(c_+ + c_-)$. Hence

$$H_{\text{merged}} = H_b(q) + qH(p^A) + (1 - q)H(p^A) \quad (19)$$

$$= H_b(q) + \log r_A. \quad (20)$$

Exponentiating proves Eq. (7). Proportional spectra have identical normalized singular-value distributions, so $r_A = r_B$. $\square$

Theorem 1. Write $E_S = \sum_{j \in S} E_j$. For PSD matrices, $\text{tr} \sqrt{A + B} \leq \text{tr} \sqrt{A} + \text{tr} \sqrt{B}$: apply the nuclear-norm triangle inequality to $[A^{1/2}, B^{1/2}]$. Hence

$$\text{tr} \sqrt{E_S} \leq \sum_j \tau_j, \quad \text{tr} \sqrt{E_S} \leq \sqrt{\text{rank}(E_S) \text{tr}(E_S)} \leq \sqrt{\min(d, \sum_j u_j) \sum_j \omega_j}, \quad (21)$$

so $\text{tr} \sqrt{E_S} \leq T_S$. Let $a_i = \sqrt{\lambda_i(\tilde{G}_S)}$ and $b_i = \sqrt{\lambda_i(G_S)}$ in decreasing order. Since $G_S \succeq \tilde{G}_S$, eigenvalue monotonicity gives $b_i \geq a_i$. With $\tilde{\nu} = \sum_i a_i$ and $\delta = \sum_i (b_i - a_i)$, trace subadditivity gives $0 \leq \delta \leq T_S$. If $\delta > 0$, the normalized true spectrum is

$$p = \frac{\tilde{\nu}}{\tilde{\nu} + \delta} \tilde{p} + \frac{\delta}{\tilde{\nu} + \delta} q, \quad q_i = \frac{b_i - a_i}{\delta}. \quad (22)$$

Thus $\text{TV}(p, \tilde{p}) \leq \delta/(\tilde{\nu} + \delta) \leq \varepsilon_S$. Fannes–Audenaert entropy continuity on a support of size at most D_S gives Eq. (14). The $\delta = 0$ and $D_S = 1$ cases are immediate. $\square$

Corollary 6. The theorem places every exact candidate score in $[\hat{F}_j - e_j, \hat{F}_j + e_j]$. Equation (15) puts the lower endpoint of j^* strictly above every competing upper endpoint. $\square$

Status of Eq. (8). For unequal or non-proportional spectra, the two branches need not inherit the input distributions, and q can vary by mode. Equation (8) therefore defines a structured approximation; no exactness or universal error bound is asserted for it.

C Optimization Limits

Effective-rank maximization is not generally monotone. Take $G_A = \text{diag}(1/2, 1/2)$ and $G_B = \text{diag}(1, 0)$. Then $\mathcal{R}(G_A) = 2$, whereas $\mathcal{R}(G_A + G_B) = 1.928623 < 2$. It is also not generally submodular. Let $G_0 = \text{diag}(1/5, 4/5)$, $G_1 = \text{diag}(0, 1)$, and $G_2 = \text{diag}(1, 0)$. For $A = \{0\} \subset B = \{0, 1\}$, direct evaluation of $F(S) = \log \mathcal{R}(\sum_{j \in S} G_j)$ gives

$$F(A \cup \{2\}) - F(A) = 0.051522 < 0.125701 = F(B \cup \{2\}) - F(B), \quad (23)$$

violating diminishing returns. Thus standard monotone-submodular greedy guarantees do not apply; exhaustive search is used only as a finite small-pool reference.

D Bootstrap Confidence Intervals

Table 4: Bootstrap 95% CIs ($B = 10,000$, seed=42) for all key metrics.

Experiment	Metric	Point estimate	95% CI	SE
E1a (280 pairs)	R^2	0.9999	[0.9999, 0.9999]	0.000013
E1a	Superadd. ($\alpha < 1$)	1.0000	[1.000, 1.000]	0.000000
E1b (30 pairs)	Pearson r	0.9478	[0.906, 0.978]	0.019
E1b	Superadd. %	0.8667	[0.733, 0.967]	0.062
E1c (57 pairs)	R^2 (global)	0.049	[-0.346, 0.287]	0.161
E1c	Pearson r (global)	0.928	[0.894, 0.949]	0.014
E1c	Superadd. %	0.930	[0.860, 0.982]	0.034
E1c (ResNet-50)	Superadd. %	1.000	[1.000, 1.000]	0.000
E1c (ResNet-50)	Pearson r	0.934	[0.885, 0.965]	0.021
E1c (ViT-B/16)	Superadd. %	0.733	[0.467, 0.933]	0.114
E1c (ViT-B/16)	Pearson r	0.945	[0.883, 0.982]	0.026
E1c (SBERT)	Superadd. %	1.000	[1.000, 1.000]	0.000

E Bimodal Principal-Angle Structure in Self-Supervised Encoders

The per-dimension $\cos^2 \theta_i$ profiles and their prediction consequences are shown in Figure 2 (main text): CLIP/DINO exhibit a two-cluster step structure (top-5 ratios $20 \times / 31 \times$ over bot-5, $V_\alpha = 0.090/0.073$), while ResNet-50 decays monotonically ($V_\alpha = 0.019$).

Two distinct mechanisms drive the low- R^2 probes. For DINO ($R^2 = 0.55$), the cause is the bimodal $\cos^2 \theta_i$ analysed below. For ViT ($R^2 = 0.49$, supervised), the cause is energy imbalance rather than bimodality: 17/17 ViT non-superadditive pairs lie in $\gamma \notin [0.5, 2]$ (cross-resolution pairs such as MNIST $\times$ ImageNet), where the global energy summary is extreme and the two-component mixture approximation degrades. ViT’s principal-angle distribution remains near-uniform (no bimodality); its lower R^2 instead empirically marks the energy-imbalance regime as a reliability boundary. We retain the full ViT row in Table 2 for transparency, flagging $\gamma \in [0.5, 2]$ as ViT’s reliable applicability window.

Honest scope. We treat the bimodal failure mode as a genuine boundary of the current formula, not a feature. The uniform- α assumption underlying The global-summary approximation is stressed whenever the principal-angle distribution is bimodal, and self-supervised contrastive/distillation encoders (CLIP, DINO) systematically produce such distributions on cross-domain pairs. Within this regime, the calibrated formula still ranks correctly (Pearson $r \geq 0.90$ on all four probes, Table 2) but the absolute prediction $\hat{r}$ is biased upward and the superadditivity-decision boundary is shifted, explaining the 29–36% observed superadditivity rates on CLIP/DINO cross-domain pairs. What the Collapse Model does contribute

here is mechanistic localisation: the bimodal top/bottom ratio $>20\times$ in $\cos^2 \theta_i$ (Figure 2) exposes alignment heterogeneity discarded by the scalar summary, rather than leaving the failure as a black-box surprise.

What does not work. A naive energy-weighted patch $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ worsens predictions (Pearson r drops 1–2%, MAE rises 6–8%), because it overweights the most-aligned direction’s redundancy while suppressing intermediate-direction contributions.

What we believe will work, and how to test it. The natural next step is a piecewise paired-direction model: partition the k directions into a high-alignment cluster ($\cos^2 \theta_i \geq \tau$) and a low-alignment cluster, apply Lemma 2 separately within each cluster, and combine the resulting branch spectra. This retains more of the distribution of α_i than a single mean, but it still requires controlling spectral pairing and cross-direction coupling. We leave this extension and its error analysis to future work.

F Historical Compression Pilot at $M=14$ Vision and $M=7$ Text Pools

This pre-audit experiment applies Collapse-4S greedy across three encoder families and two modalities, using FacLoc / k -Medoids as baselines. It predates the label-blind cache, source weighting, and Collapse-Gram- L implementation; all numbers below are descriptive historical results.

Table 5: Diversity retention $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%) across strategies, architectures, and modalities. FacLoc = Facility Location (submodular greedy on SA_k kernel); kMedoids = PAM on $1-\text{SA}_k$ distance. Random is the 10-seed mean $\pm$ std. The table evaluates spectral retention, not a task-utility guarantee.

Probe	M	k (ratio)	Collapse-4S	FacLoc	kMedoids	Random	LP _{4S}
Vision							
ResNet-50	14	3 (4.7 $\times$)	101.4	87.8	87.5	84.3 $\pm$ 10.0	0.754
ResNet-50	14	5 (2.8 $\times$)	105.7	95.6	92.6	88.7 $\pm$ 5.3	0.754
ResNet-50	14	7 (2.0 $\times$)	105.9	98.4	98.0	93.1 $\pm$ 7.6	0.747
ViT-B/16	11	3 (3.7 $\times$)	103.6	—	—	92.0 $\pm$ 3.3	0.779
ViT-B/16	11	5 (2.2 $\times$)	104.9	—	—	94.1 $\pm$ 3.7	0.779
Text							
BERT L6	7	3 (2.3 $\times$)	103.6	—	—	97.8 $\pm$ 5.0	—
RoBERTa L6	7	3 (2.3 $\times$)	103.8	—	—	97.0 $\pm$ 6.1	—
GPT-2 L6	7	3 (2.3 $\times$)	104.8	—	—	92.8 $\pm$ 9.5	—
GPT-2 L6	7	5 (1.4 $\times$)	106.0	—	—	96.7 $\pm$ 4.7	—

On this ResNet-50 pool, Collapse-4S has 7.5–13.6 percentage points higher spectral retention than Facility Location and 7.9–13.9 points higher than k -Medoids. The accompanying LP values are a narrow pilot and were not produced by the corrected two-stage protocol; they do not establish that the retention ordering transfers to new targets.

G r_A+r_B Heuristic: Safe Regime and Failure Mode

When does r_A+r_B suffice, and when does it fail? On clean curated pools, the r_A+r_B and Max- r_{eff} heuristics often select the same subset as Collapse-4S (within 0.7 pp in the historical $M=43$ stress test). This agreement is consistent with the summary diagnostic in Corollary 4: when all top- r_{eff} candidates have low mutual alignment ($\bar{\alpha} < 0.2$, i.e. rank-ratio $r_{\text{dom}}/r_{\text{sub}} < \rho_c$), the superadditivity bonus $H_b(q)$ is dominated by the rank term $q \log r_{\text{dom}} + (1-q) \log r_{\text{sub}}$, making individual rank a near-sufficient statistic. Collapse-4S thus serves a different role than “beating r_A+r_B everywhere”: (i) A priori low-redundancy diagnostic. $\rho_c(\bar{\alpha}, \bar{\gamma})$ computed from per-dataset summaries indicates when r_A+r_B is likely to agree without actually running competing strategies. On our ResNet-50 vision pools, $\bar{\alpha} < 0.2$ between top- r_{eff} pairs places $r_{\text{dom}}/r_{\text{sub}}$ well inside the predicted agreement regime ($\rho_c > 4$ at $\bar{\alpha} \leq 0.2$, Cor. 4), matching the observed heuristic-equivalence at $M=14$ and $M=43$. (ii) Detection of the unsafe regime. When candidates contain near-duplicate or strongly aligned

subspaces, the α channel supplies information that rank-only omits. The clone experiment below (r_A+r_B retention drops 25.5%, Collapse holds) demonstrates this on a controlled stress test; the naturally bimodal CLIP/DINO cross-domain pairs (Sec. 4.2, $\alpha \in [0.35, 0.41]$ with top-5 $\cos^2 \theta \geq 0.7$) are real-world settings where the α signal is non-negligible and the rank-only heuristic can fail. (iii) Diagnosis without execution. The Collapse-4S regime classification is computable from (γ, α) , but Proposition 5 makes it a heuristic diagnostic, not a certified decision rule.

Redundancy clone trap: failure mode of r_A+r_B . To stress-test the alignment term under constructed redundancy, we construct a clone attack scenario: the highest- r_{eff} dataset in the pool (STL-10, $r_{\text{eff}}=994$) is cloned via random rotation $H_{\text{clone}}=H_{\text{orig}}Q(\theta)$ ($\theta \in \{0.01, 0.03, 0.05, 0.07, 0.10\}$ radians), producing 5 clones that preserve the original r_{eff} but are highly aligned with the original ($\text{SA}_k > 0.99$). On a pool of 5 real + 5 cloned ResNet-50 datasets, r_A+r_B greedy at $k=3$ selects 2/3 clones ($r_{\text{eff}}^{\text{merged}}=997$), while Collapse-4S assigns a redundancy penalty via high α , selecting 0/3 clones ($r_{\text{eff}}^{\text{merged}}=1251$, +25.5%). At $k=5$ the gap persists: r_A+r_B selects 4/5 clones ($r_{\text{eff}}=998$), Collapse only 2/5 ($r_{\text{eff}}=1233$). This result is consistent across all 5 θ values ($\theta=0.01-0.10$, SA_k from 0.9999 to 0.9891). Because the duplicated source was deliberately chosen from the highest-rank candidate, this is an adversarial stress test rather than evidence about natural-pool frequency. It shows only that alignment can help in this constructed case.

H P2-E: Hybrid Two-Signal Strategy

The historical $M=40$ pilot suggested a two-signal hybrid: seed $\lceil k/2 \rceil$ datasets via one signal, fill the remaining $\lfloor k/2 \rfloor$ via the other. We test three orderings on that same $M=40$ ResNet-50 pool: Col→Cen (Collapse first, Centroid-FF on residuals), Cen→Col (Centroid-FF first, Collapse on residuals), and Col→Dom (Collapse first, DomainDiverse on residuals).

Table 6: Two-signal hybrid strategies on $M=40$ ResNet-50. Pure-strategy reference rows are shown for comparison. Bold marks the best LP at each k .

Strategy	$k=5$ (8.0×)		$k=7$ (5.7×)		$k=10$ (4.0×)	
	Ret%	LP	Ret%	LP	Ret%	LP
Pure (reference)						
Collapse	117.2	0.7576	117.4	0.7578	116.2	0.7576
Centroid-FF	93.4	0.7570	96.7	0.7624	100.5	0.7630
DomainDiverse	114.7	0.7634	111.8	0.7596	113.8	0.7544
Hybrid two-stage						
Col→Cen	110.2	0.7534	111.9	0.7496	113.8	0.7546
Cen→Col	106.7	0.7596	113.6	0.7610	115.0	0.7602
Col→Dom	114.7	0.7634	116.0	0.7562	113.8	0.7544

Historical result. At $k \in \{7, 10\}$ (the operational compression range), Cen→Col achieves LP 0.7610/0.7602, exceeding pure Collapse (0.7578/0.7576) by +0.0032/+0.0026 while sacrificing only 3.8/1.2 pp of retention. At $k=10$, Cen→Col trades 1.2 percentage points of retention for 0.0026 higher LP; it does not dominate pure Collapse on both metrics. Compared to pure Centroid-FF, Cen→Col recovers 13.3 pp of retention at $k=5$, 16.9 pp at $k=7$, and 14.5 pp at $k=10$ while keeping LP within ± 0.0028 . Col→Cen, in contrast, hurts both metrics: once Collapse has selected high- r_{eff} datasets, adding "different prototype" datasets dilutes the spectrum without recovering centroid coverage. Col→Dom reduces to DomainDiverse on this pool because DomainDiverse already picks STL-10 and fashion_mnist (the highest- r_{eff} datasets in their respective domains) that Collapse would also select.

Implication. The ordering matters: starting with Centroid-FF establishes a geometric backbone that Collapse then augments with high- r_{eff} complementary directions. The summary formula's role in this hybrid is to fill the residual budget with diversity-maximizing choices, exactly the regime where it operates with summary statistics alone (the centroid-FF prefix can be precomputed once from already-shared centroids, no feature-matrix exchange

needed). This single-pool result motivates, but does not validate, a composable two-signal strategy. It must be repeated with the corrected protocol and independent targets.

I E5 Supplementary: Per-Target Selection Details

Per-target regret table ($M=10$ pool).

Strategy	Mean regret (%) / Mean recall (%)			
	$K=1$	$K=2$	$K=3$	$K=5$
Collapse-4S	0.0 / 100	0.0 / 100	0.0 / 100	1.0 / 87
Max- r_{eff}	0.0 / 67	2.0 / 67	0.9 / 89	1.0 / 87
Δ_{dir} -greedy	20.3 / 0	16.7 / 17	17.0 / 33	13.3 / 40
Random	15.2 / -	17.5 / -	16.1 / -	11.1 / -

For BloodMNIST ($K=1$), Max- r_{eff} selects STL-10 (highest individual $r_{\text{eff}}=357$) but Collapse-4S selects CIFAR-100 ($r_{\text{eff}}=356$, marginally lower) because the latter has lower alignment with BloodMNIST ($\alpha=0.18$ vs. 0.21), yielding a higher predicted merged rank—confirmed by $r_{\text{eff}}^{\text{merged}} = 506.8$ vs. 506.4.

r_A+r_B greedy baseline. We evaluate an r_A+r_B greedy strategy that selects the candidate with the highest individual r_{eff} at each round. On the $M=10$ pool, this baseline matches Collapse-4S at $K \leq 3$ (identical subset, 0% regret), diverging only at $K=5$ where Collapse-4S retains 100% recall while r_A+r_B drops to 87%. This mirrors the ablation finding: r_A+r_B suffices when candidates have distinct individual ranks, but fails when alignment discrimination (α) is needed.

Downstream LP validation. We train a linear probe on the merged features from each strategy’s K -subset and evaluate on held-out test data. All four strategies (Exhaustive, Collapse-4S, r_A+r_B , Max- r_{eff}) produce identical LP accuracy across all 6 tested configurations (CIFAR-10 and STL-10 targets, $K \in \{1, 2, 3\}$), ranging from 0.776 (CIFAR-10, $K=3$) to 0.979 (STL-10, $K=1$). Because every method has identical LP accuracy here, this experiment does not distinguish selectors and cannot show that maximizing merged r_{eff} improves downstream utility.

J E7 Supplementary: kNN Downstream Validation

The main text uses linear-probe (LP) accuracy as one downstream metric. Here we compare whether k -nearest-neighbour (kNN) gives similar rankings on the evaluated configurations; this cannot establish LP as sufficient for arbitrary downstream tasks.

Part B: LP-kNN rank correlation. We compute LP, kNN-5, and kNN-20 accuracy on 5 single datasets (CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST) across 11 encoder families (ResNet-50/101, ViT-B/16, DeiT-B, Swin-Tiny, BEiT-Base, ConvNeXt-Tiny, DenseNet-121, EfficientNet-B4, MaxViT-Tiny, RegNet-Y), yielding 53 observations. Spearman rank correlation: $\rho(\text{LP}, \text{kNN-5})=0.980$, $\rho(\text{LP}, \text{kNN-20})=0.973$ —LP and kNN rankings are near-identical.

Part A: compression advantage under kNN. Using the $M=6$ ResNet-50 pool from Sec. 5, we evaluate Collapse Greedy vs. 10 random seeds using LP, kNN-5, and kNN-20 on the merged features (averaged over 6 targets).

k	Strategy	LP	kNN-5	kNN-20
3	Collapse	73.5	62.4	61.2
	Random (μ)	72.6	58.2	53.8
	Δ	+0.9	+4.2	+7.4
5	Collapse	71.5	57.8	53.5
	Random (μ)	71.1	56.8	51.1
	Δ	+0.4	+1.0	+2.4

On this historical $M=6$ pool, the Collapse-4S-random difference is larger for kNN than for LP; at $k=3$, for example, it is 7.4 percentage points for kNN-20 and 0.9 for LP. The high single-dataset LP-kNN correlation ($\rho>0.97$) and the strategy comparison reuse a small, shared pool, so we treat LP only as one proxy rather than a sufficient one.

K Additional Notes

Note on multiple comparisons. The within-probe bootstrap analyses (Sec. 5) involve 11 probes tested independently. Since the primary claim is about the average $\Delta\rho$ across probes (not individual-probe significance), Bonferroni correction is not required for the aggregate Fisher z -test ($p=0.74$). For the vision-only bootstrap ($\Delta\rho= +0.005$, 95% CI $[+0.001, +0.011]$), a conservative Bonferroni threshold of $\alpha=0.05/11=0.0045$ would still be met ($P(\text{Collapse} > r_sum)=0.99$).

L Scope of the Paired-Direction Model

Remark 1 (Exact and approximate layers). Proposition 1 is unconditional and retains the full coupling matrix $C = Q_A^\top V_B$. Lemma 2 is exact only when the singular directions can be paired into mutually orthogonal two-dimensional planes. Proposition 3 additionally requires mode-wise proportional spectra and uniform alignment; these conditions imply $r_A = r_B$. The practical predictor Eq. (8) allows $r_A \neq r_B$ by interpolating two input entropies, and is therefore a modeling approximation rather than a consequence of the proportional-spectrum proposition.

Remark 2 (Information discarded by global summaries). The global ratio $\gamma = \|H_B\|_* / \|H_A\|_*$ does not imply $\sigma_{B,i} = \gamma\sigma_{A,i}$. Likewise, $\alpha = \text{SA}_k$ does not retain the distribution of α_i , its association with spectral mass, or dense cross-direction coupling. Proposition 5 gives an explicit pair of examples with identical $(r_A, r_B, \gamma, \alpha)$ and different merged effective ranks. Consequently no estimator based only on these four scalars can be universally exact.

Remark 3 (Empirical generalization). The summaries in Eq. (8) remain well-defined outside the paired-direction model. Their empirical correlations on synthetic out-of-model generators and real encoders quantify predictive utility, not theorem validity. Alignment heterogeneity, non-proportional spectra, and cross-direction coupling are observable failure sources; calibration corrects average scale on the evaluated encoder family but does not create a general guarantee.

Remark 4 (Calibrated Collapse Model). On real data the paired-direction summary model is approximate—cross-pair coupling, non-proportional spectra ($\sigma_{B,i} \neq \gamma\sigma_{A,i}$, hence $r_A \neq r_B$), and higher-order interactions introduce a systematic upward bias on the evaluated pairs. Empirically the over-prediction ratio is $1.24\text{--}1.84\times$ (E1c), and its gap is positively associated with alignment heterogeneity. Since the log-space bias has low variance (std = 0.164 vs. mean = 0.293), a single multiplicative calibration restores quantitative accuracy:

$$\hat{r}_{\text{cal}} = c \cdot \hat{r}_{\text{merged}}, \quad c = \exp\left(-\frac{1}{n} \sum_{i=1}^n \log \frac{\hat{r}_i}{r_i}\right). \quad (24)$$

This calibration compensates for the entropy over-estimation caused by structure that the four global summaries ignore. With $c=0.746$ estimated from E1c ($n=57$), LOO-CV yields $R^2=0.855$, RMSE=87.7 (vs. uncalibrated $R^2=0.049$, RMSE=224; Sec. M.7). The constant generalizes across datasets within each encoder family: calibrating on the 30 E1b pairs ($c=0.768$) and testing on the held-out 57 E1c pairs yields $R^2=0.859$, RMSE=86.3 (Table 11). However, c varies across encoder families ($c=0.78$ for ResNet-50 vs. $c=0.65$ for ViT-B/16), reflecting systematic differences in how each architecture’s spectral structure deviates from the paired-direction summary model. On the evaluated encoder families, this variation is fit by:

$$c = 0.946 - 0.175 \log(N/r_{\text{eff}}), \quad R^2=0.963, \quad \text{LOO mean error}=0.027, \quad (25)$$

This small-sample regression is descriptive and requires validation on a new encoder family before use; its in-sample R^2 is not evidence of a universal law.

650 L.1 Why a Task-Aware Second Stage Is Necessary

651 Unlabeled Grams contain no information about an arbitrary target labeling. Indeed, two
652 tasks may share exactly the same frozen features while assigning labels to different directions,
653 reversing the utility order of two source subsets. Therefore no label-free spectral score can
654 universally guarantee downstream utility. Our Stage-2 validation is not a consequence of
655 Theorem 1: it is the separate empirical step needed to measure target-conditioned utility
656 under equal training compute.

Algorithm 1 Collapse-Gram- L Greedy Screening

Require: Summaries $\{(B_j, \tau_j, \omega_j, u_j)\}_{j=1}^M$, budget K
Ensure: Ordered shortlist S , with a certificate flag for each step
1: $S \leftarrow \emptyset$; $C \leftarrow \{1, \dots, M\}$
2: for $t = 1, \dots, K$ do
3: for each $j \in C$ do
4: compute $\hat{F}_j$ and e_j for $S \cup \{j\}$ using Eqs. (11)–(14)
5: end for
6: $j^* \leftarrow \arg \max_{j \in C} \hat{F}_j$
7: $z_t \leftarrow [\hat{F}_{j^*} - e_{j^*} > \max_{j \neq j^*} (\hat{F}_j + e_j)]$
8: $S \leftarrow S \parallel j^*$; $C \leftarrow C \setminus \{j^*\}$
9: end for
10: return $(S, z_1, \dots, z_K)$

657 Certified Stage-1 pseudocode.

658 M Additional Experiments (E1a, E1a-Ind, E1b, E1c, Calibration,
659 E_diverse, E4, E_downstream)

660 M.1 Ranking Ablation Table

Table 7: Ranking ablation: Spearman ρ between predictor and true merged r_{eff} (256 vision pairs, 4 encoder families).

Predictor	Channels used	ResNet-50 ($n=91$)	ViT-B/16 ($n=55$)	All vision ($n=256$)
Collapse Model	$\Sigma+V+U$	0.896	0.923	0.966
$r_A + r_B$	U only	0.892	0.946	0.963
$\max(r_A, r_B)$	U only	0.898	0.779	0.953
$1 - \alpha$ (V only)	V only	−0.289	−0.192	0.450
γ (Σ only)	Σ only	0.277	0.001	0.271

661 M.2 E1a: Synthetic Controlled Validation

662 The E1a synthetic validation results (Table 1) are presented in the main text; Figure 3
663 visualizes the (α, γ) landscape and superadditivity region. Additional design details:
664 feature pairs are generated from the proportional-spectrum paired-direction model with
665 $\alpha \in \{0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and $\gamma \in \{0.1, 0.33, 0.5, 1.0, 2.0, 3.0, 10.0\}$, $d = 512$,
666 $N = 1000$, 5 independent repeats per setting (280 pairs). All 35 non-superadditive cases
667 concentrate exclusively at $\alpha = 1$ (identical subspaces), confirming the degenerate boundary
668 of Corollary 4.

669 M.3 E1a-Ind: Independent Validation on Non-Principal-Angle Matrices

670 Motivation and design. A critical concern with E1a is circularity: synthetic data gener-
671 ated from the proportional-spectrum paired-direction model satisfies the exact model as-
672 sumptions, making $R^2 = 0.9999$ a near-tautological result. To independently test whether

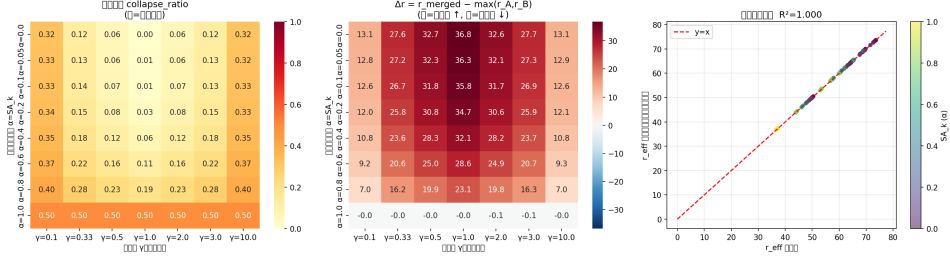


Figure 3: E1a: Collapse Model prediction accuracy on the synthetic (α, γ) grid. Left: collapse ratio heatmap. Center: superadditivity ($\delta_r > 0$) region. Right: predicted vs. actual r_{eff} scatter ($R^2=0.9999$).

Table 8: E1a-Ind: Independent validation on 120 non-principal-angle synthetic pairs. Pearson r measures directional ranking; $R^2 < 0$ indicates systematic over-prediction on the observed generator. SA% = fraction of pairs where the formula predicts superadditivity.

Generation Method	n pairs	Pearson r	R^2	SA%
M1: Low-rank Gaussian	30	0.9985	+0.9949	100%
M2: Shared+private subspace	30	0.9982	+0.9963	100%
M3: Sparse activations	30	0.9991	−14.65	100%
M4: Block-diagonal	30	0.9996	−12.32	100%
Overall	120	0.9987	−1.09	100%

Eq. (8) remains predictive beyond its derivation assumptions, we generate 120 synthetic matrix pairs using four generation methods that do not use that paired model: (M1) low-rank Gaussian: $H = U_k \Sigma V_k^T + \epsilon$ with random orthogonal bases; (M2) shared+private subspace: $H = H_{\text{shared}} + H_{\text{private}}$ with partial overlap enforced by construction; (M3) sparse activations: rows drawn from k -sparse Gaussian mixtures; (M4) block-diagonal: features from two independent subspaces with no principal-angle structure. $d = 512$, $N \in \{500, 1000\}$; 30 pairs per method.

Results. Table 8 shows Pearson $r = 0.9987$ overall: the formula correctly ranks all 120 pairs across all four generation methods. R^2 is method-dependent: ≥ 0.99 for Gaussian-structure methods (M1, M2), but strongly negative for sparse and block-diagonal structures (M3, M4), because those structures systematically reduce actual r_{eff} relative to the formula’s prediction. This reveals excellent ordering but poor absolute prediction on sparse and block-diagonal structures, exposing structural sparsity as an empirical failure mode. Superadditivity is predicted at 100% for all methods.

Remark 5 (Formula as directional oracle, not absolute predictor). The contrast between $R^2 = -1.09$ (overall, E1a-Ind) and Pearson $r = 0.9987$ (same data) is not contradictory: R^2 measures absolute calibration, while Pearson r measures directional ranking. This distinction is crucial for the real-data results (E1c): the formula yields $R^2 \approx 0.05$ (poor absolute calibration, 1.24–1.84 $\times$ over-prediction) but Pearson $r = 0.928$ (correct directional ranking). For practical applications—ranking candidate merges by diversity gain, dataset selection, data valuation—ranking is the operationally relevant metric, not absolute r_{eff} prediction. Absolute calibration requires a correction factor (calibrated $\hat{r} = 0.81 \cdot \hat{r}_{\text{pred}}$, $R^2_{\text{cal}} = 0.862$ on E1c data), which we report as a practical recommendation.

M.4 E1b: Real Dataset Pairs (30 Pairs)

Design. We evaluate all $\binom{6}{2} = 15$ pairs from $\{\text{CIFAR-10, STL-10, SVHN, MNIST, Fashion-MNIST, CIFAR-100}\}$ using 2 pretrained probes (ResNet-50 [He et al., 2016] $d=2048$ and ViT-B/16 [Dosovitskiy et al., 2021] $d=768$), $N_{\text{sub}} = 500$ samples per dataset: 30 total configurations. The Collapse Model is applied with per-probe nuclear-norm γ .

Results. 86.7% of pairs are superadditive (26/30; 95% CI [73%, 97%]). Pearson $r = 0.948$ (95% CI [0.906, 0.978]) between predicted and actual merged rank. Per-probe: ResNet-50 achieves 100% superadditivity (15/15, Pearson $r = 0.968$); ViT-B/16 achieves 73.3% (11/15, Pearson $r = 0.945$).

All 4 exceptions are ViT-B/16 pairs involving CIFAR-100. The exceptions are associated with energy imbalance, consistent with the superadditivity diagnostic (Corollary 4). CIFAR-100 has 100 classes and correspondingly high intrinsic complexity: $r_{\text{eff}}(\text{CIFAR-100}) \approx 526$ under ViT-B/16, far exceeding partner datasets such as MNIST ($r_{\text{eff}} \approx 140$) and SVHN (≈ 243). This yields energy ratios $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary $\rho_c(\alpha, \gamma)$. Specifically, the dominant component absorbs the subordinate: $q \rightarrow 1$, $H_b(q) \rightarrow 0$, and $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$, so the observed r_{merged} can fall slightly below r_{dom} due to spectral averaging. We verify this by sweeping $N_{\text{sub}} \in \{500, 1000, 2000, 4000\}$: at $N=500$, CIFAR-100’s r_{eff} is compressed to ≈ 343 (comparable to partners), giving $\gamma \approx 1$ and 100% superadditivity (5/5); as N grows to 2000+, r_{eff} stabilizes at 526, γ increases, and exceptions emerge. These quantities co-vary and are consistent with the energy-imbalance explanation, but the sweep does not identify it causally. ResNet-50 ($d=2048$, $r_{\text{eff}} \approx 988$ for CIFAR-100) yields even larger γ ratios, yet remains 100% superadditive (36/36) because its partner datasets also have higher r_{eff} in 2048-dimensional space, keeping $r_{\text{dom}}/r_{\text{sub}}$ below ρ_c .

M.5 E1c: Large-Scale Multi-Domain Validation (57 Pairs)

Design. To test generalization beyond vision pairs, we extend to three probes and three domain types: ResNet-50 on 9 datasets (36 pairs, natural+medical), ViT-B/16 on 6 datasets (15 pairs, natural), and SBERT [Reimers and Gurevych, 2019] on 4 text domains (6 pairs, text). Total: 57 pairs, 3 probes, 3 domain types. $N_{\text{sub}} = 2000$ for natural/medical; $N_{\text{sub}} = 1000$ for text.

Results. Table 9 shows strong and consistent superadditivity across all probes and domains.

Table 9: E1c: Historical Collapse-4S association study. Superadditivity rate and Pearson r (predicted vs. actual merged r_{eff}). Bootstrap 95% CI ($B = 10,000$). The formula over-predicts on all probes in this experiment; “Bias” reports the observed multiplicative ratio.

Probe / Domain	n	Superadd. % (95% CI)	Pearson r (95% CI)	Bias
ResNet-50 (natural+medical)	36	100% [100,100]	0.934 [0.885,0.965]	1.24×
ViT-B/16 (natural)	15	73.3% [46.7,93.3]	0.945 [0.883,0.982]	1.45×
SBERT (text)	6	100% [100,100]	0.780	1.84×
Overall	57	93.0% [86.0,98.2]	0.928 [0.894,0.949]	1.37×

Key findings. (i) 93% superadditivity on this mixed benchmark is descriptive; the six SBERT pairs are too few to establish domain- or modality-agnostic behavior. (ii) All 4 exceptions are ViT-B/16 pairs involving CIFAR-100, attributable to energy imbalance (same mechanism as E1b, Sec. M.4): CIFAR-100’s high intrinsic complexity ($r_{\text{eff}} \approx 526$, 100 classes) creates $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary ρ_c of Corollary 4. (iii) Pearson $r = 0.928$ vs. $R^2 \approx 0.05$: these metrics measure different properties. The formula over-predicts by 1.24–1.84× on these probes, so R^2 is low before calibration; Pearson r measures ordering. A single-parameter calibration (Eq. 24) restores quantitative accuracy: $R^2_{\text{cal}} = 0.855$ (LOO-CV), and cross-validation from E1b to E1c yields $R^2 = 0.859$ (see Sec. M.7 and Remark 4). (iv) ResNet-50: 100% superadditivity (36/36) is strong empirical agreement with the summary diagnostic on this observed set. (v) Over-prediction scales with n/r_{eff} : ResNet-50 ($n = 2000 \gg r_{\text{eff}} \approx 350$, 1.24×) < ViT-B/16 ($n = 2000 \approx 3r_{\text{eff}} \approx 650$, 1.45×) < SBERT ($n = 1000 \approx r_{\text{eff}} \approx 700$, 1.84×). This is an observed trend in a small set, not a finite-sample theorem.

M.6 E1d: Large-Scale Validation (427 Pairs)

Design. To test the Collapse Model at scale and on ImageNet-derived features, we construct a comprehensive evaluation spanning 5 encoder families (ResNet-50, ViT-B/16, CLIP

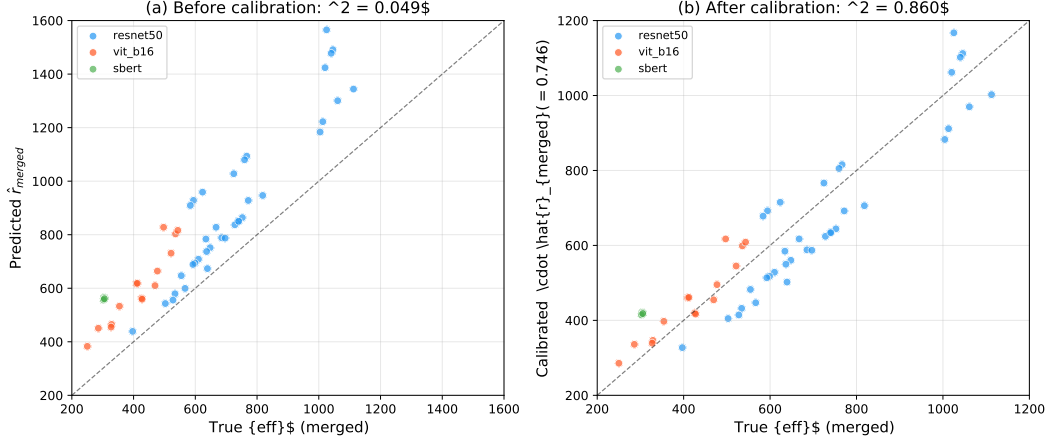


Figure 4: E1c: Predicted vs. actual merged r_{eff} for 57 real-data pairs (5 probes, 3 domain types). (a) Before calibration: systematic over-prediction ($R^2=0.049$). (b) After single-parameter calibration ($c=0.746$): $R^2=0.860$. Pearson $r=0.928$ is identical in both panels—calibration corrects scale, not ranking.

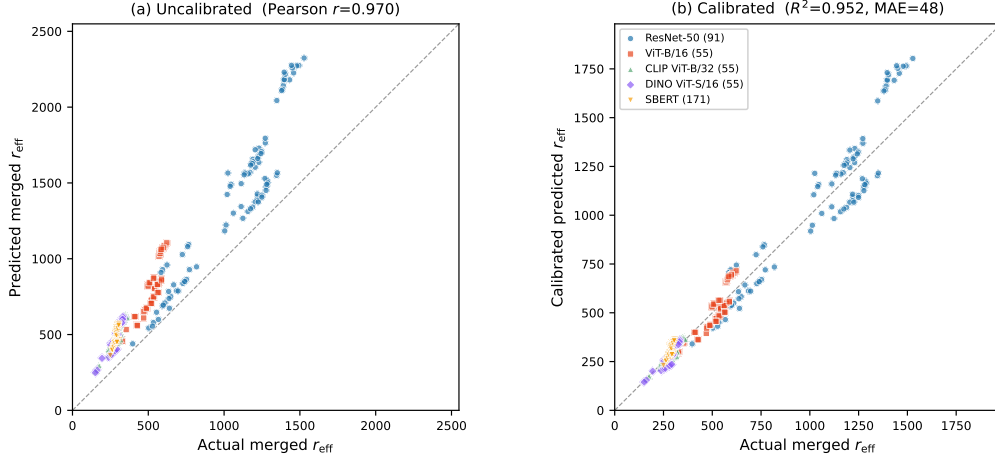


Figure 5: Predicted vs. actual merged r_{eff} for all 427 pairs across 5 encoder families. (a) Uncalibrated: predictions are systematically high but strongly correlated (Pearson $r=0.970$). (b) After per-probe calibration: $R^2=0.952$, MAE=48. Color encodes encoder family; the identity line is dashed.

743 ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN,
 744 MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet se-
 745 mantic subsets: animal, vehicle, artifact, food, scene), and 19 SBERT text corpora (AG
 746 News, 20 Newsgroups, DBpedia, IMDB, Yelp subsets). Total: 427 pairs (91 ResNet-50, 55
 747 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT). The 5 ImageNet subsets are constructed from
 748 ImageNet-1K validation images grouped by WordNet semantic hierarchy, with $N=2000$
 749 samples each.

750 Results.

751 Key findings. (i) At 427 evaluated pairs the overall Pearson correlation is $r=0.970$, com-
 752 pared with $r=0.928$ in E1c. Because pairs share datasets and the larger set is not an
 753 independent replication, this increase does not validate a theorem. (ii) All 20 ImageNet-
 754 only pairs are superadditive, and all 30 natural×ImageNet cross-domain pairs are superad-
 755 ditive; this is an empirical property of these extracted features. (iii) ResNet-50 achieves
 756 100% superadditivity across 91 pairs (14 datasets including medical and ImageNet) on

Table 10: E1d: Historical Collapse-4S association study (427 pairs, 5 probes).

Probe	n	Superadd. %	Pearson r	c	R_{cal}^2
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	per-probe	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	per-probe	0.952

this benchmark. In contrast, CLIP ViT-B/32 and DINO ViT-S/16 achieve only 31% and 38% superadditivity respectively. Analysis reveals a clear pattern: within-domain pairs (natural×natural, ImageNet×ImageNet) achieve near-perfect superadditivity (CLIP: 90% on ImageNet-only, 100% on natural-only; DINO: 100% on both), while cross-domain pairs (handwritten×ImageNet, natural×digit) fail for nearly all evaluated pairs. This suggests that self-supervised encoders learn domain-specific spectral structures with heterogeneous alignment and spectral coupling discarded by the global summaries—an important boundary condition for the Collapse Model’s applicability. (iv) On 171 SBERT text pairs the correlation drops to $r=0.793$ (98.8% superadditive). The narrow r_{eff} range (IQR: 285–298) may contribute, but these results do not establish cross-modal generalization. (v) Calibration constant varies by encoder family ($c \in [0.57, 0.78]$), but a single global $c=0.712$ achieves $R^2=0.898$ across all 158 vision pairs, describing the average bias on this benchmark.

Understanding non-superadditive pairs. ViT-B/16 achieves only 69% superadditivity (38/55), lower than ResNet-50’s 100% (91/91). Analysis of the 17 ViT failures reveals a clear pattern: all failures occur in the energy-imbalance regime ($\gamma \gg 1$ or $\gamma \ll 1$). At $\gamma \geq 3$, 80% of pairs fail (8/10); at $\gamma \in [0.5, 1.5]$, only 4% fail (1/24). The failures concentrate in handwritten×ImageNet cross-domain pairs (10/10 fail, 100%)—these combine low-rank handwritten features ($r_{\text{eff}} \approx 140$ for MNIST via ViT-B/16) with high-rank ImageNet features ($r_{\text{eff}} \approx 600$), creating $\gamma \in [2.0, 4.6]$ far beyond the balanced regime where the two-component spectral mixture is accurate. At large scale (55 pairs each), CLIP ViT-B/32 and DINO ViT-S/16 show only 29% and 36% superadditivity overall, with failures concentrating in cross-domain pairs: handwritten×ImageNet (100% fail), natural×digit (100% fail), natural×handwritten (100% fail). Within-domain pairs (natural-only, ImageNet-only) remain near-perfect.

Root cause: bimodal principal angles. Comparing the same dataset pair across encoders reveals that the failure is not caused by energy imbalance (γ) but by a structural violation of the uniform- α assumption. Three mechanisms interact:

(a) Systematically higher alignment. On the same pair (e.g., CIFAR-10×ImageNet-animal), CLIP’s subspace alignment $\alpha=0.41$ is $2.7\times$ higher than ResNet-50’s $\alpha=0.15$. Contrastive pre-training (CLIP) and self-distillation (DINO) push cross-domain features into a shared semantic subspace, creating high directional overlap where ResNet-50 preserves domain-specific directions.

(b) Extreme spectral concentration. CLIP/DINO feature spectra are highly non-uniform: $\kappa=15,000\text{--}1,200,000$ vs. ResNet-50’s $\kappa=70\text{--}400$. This concentration makes average alignment less informative because overlap on a few high-mass directions can dominate many complementary tail directions.

(c) Bimodal principal-angle distribution. The individual principal angles $\cos^2 \theta_i$ are not approximately uniform (as the model assumes) but bimodal: the top-5 directions are nearly aligned ($\overline{\cos^2 \theta} \approx 0.94$ for CLIP on CIFAR-10×STL-10) while the bottom directions remain orthogonal ($\overline{\cos^2 \theta} \approx 0.01$). The Collapse Model averages these into $\bar{\alpha}=0.52$, predicting moderate superadditivity. In reality, the dominant (high- σ) directions are fully redundant and the orthogonal (low- σ) directions carry negligible energy, so the merge provides almost no diversity gain ($\Delta r_{\text{eff}}=31$ vs. ResNet-50’s $\Delta r_{\text{eff}}=282$ on the same pair).

This analysis identifies a precise applicability boundary: the Collapse Model is reliable when the principal-angle distribution is approximately uniform (supervised encoders, within-domain pairs), but fails when it is bimodal (self-supervised encoders on cross-domain pairs). A natural fix—replacing the uniform average $\bar{\alpha} = \text{SA}_k$ with an energy-weighted alignment $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ —would down-weight orthogonal low-energy directions and up-weight the redundant high-energy ones, potentially restoring predictive accuracy for self-supervised encoders. However, our experimental validation shows that this correction does not improve predictive accuracy: on CLIP ($n=61$), the uniform SA_k yields Pearson $r=0.951$, MAE=206, while $\bar{\alpha}_w$ yields $r=0.935$, MAE=218 (worse); DINO shows a similar pattern (r drops from 0.912 to 0.904, MAE increases by 4%); ResNet-50 and ViT-B/16 also show no improvement. A possible explanation is that simple energy weighting over-emphasises the first singular direction’s redundancy ($\cos^2 \theta_1 \approx 0.87$ for that direction), while improperly suppressing the estimated contribution of intermediate directions to the merge. More refined corrections (e.g., piecewise paired-direction models, per-direction threshold truncation, or adaptive k selection) may break through this bottleneck and are left to future work.

M.7 Calibration: From Systematic Bias to Quantitative Prediction

The Collapse Model systematically over-predicts on the evaluated real-data pairs (Remark 4). The log-space over-prediction is approximately constant on the evaluated probes and domains (mean=0.293, std=0.164), so we test whether a multiplicative constant c (Eq. 24) improves within-benchmark calibration. This does not guarantee out-of-encoder quantitative prediction.

Cross-dataset generalization. To verify that c is not dataset-specific, we perform held-out calibration: estimate c on one experiment and test on another (Table 11).

Table 11: Cross-dataset calibration. c is estimated on the calibration set and applied (without re-fitting) to the test set. All setups dramatically improve over the uncalibrated baseline ($R^2 < 0.44$).

Calibration set	Test set	c	n_{test}	R^2	RMSE
E1b (30 pairs)	E1c (57 pairs)	0.768	57	0.859	86.3
E1c (57 pairs)	E1b (30 pairs)	0.746	30	0.888	80.7
E1c (57 pairs)	E_diverse (30 pairs)	0.746	30	0.897	82.8
E1b (30 pairs)	E_diverse (30 pairs)	0.768	30	0.894	83.7
LOO-CV (all 111 pairs, original)		0.736	111	0.877	86.1
LOO-CV (158 vision pairs, E1d)		0.712	158	0.896	121.7
Global (all 427 pairs, E1d)		0.650	427	0.920	105.0

Per-probe calibration (Table 12). The calibration constant varies systematically across encoder families, reflecting differences in spectral structure:

Table 12: Per-probe calibration on the 427-pair dataset (E1d). Each probe is calibrated independently. The variation in c is monotone with the encoder’s effective spectral flatness.

Probe	n	Pearson r	c	R_{cal}^2	Superadd. %
ResNet-50	91	0.940	0.776	0.760	100%
ViT-B/16	55	0.898	0.647	0.487	69%
CLIP ViT-B/32	55	0.950	0.595	0.839	29%
DINO ViT-S/16	55	0.913	0.584	0.554	36%
Vision (global)	256	0.965	per-probe	0.942	65%

Encoders with higher effective rank relative to sample size require stronger correction (c further from 1), consistent with the finite-sample upper-bound mechanism: ResNet-50 features ($d=2048$, $r_{\text{eff}} \approx 350$) are spectrally sparser than ViT features ($d=768$, $r_{\text{eff}} \approx 400$), so the two-component spectral mixture approximation incurs less error. A global $c=0.712$ across all vision probes yields $R^2=0.898$ on the same benchmark; out-of-family validation is required before using a shared constant.

Figure 4 shows the E1c scatter plot before and after calibration.

M.8 E_diverse: Cross-Architecture Probe Independence

Design. To test whether the Collapse Model’s predictions are robust to the choice of encoder, we evaluate the same 6 dataset pairs (from E1b) using both ResNet-50 and ViT-B/16 as independent probes, giving 12 total configurations.

Results. ResNet-50: $R^2 = 0.557$, Pearson $r = 0.9987$, 100% superadditive. ViT-B/16: $R^2 = -16.1$ (systematic over-prediction), Pearson $r = 0.9825$, 83.3% superadditive. Cross-architecture Pearson $r \geq 0.982$ on these six pairs suggests similar directional rankings, but cannot establish probe independence. The Pearson vs. R^2 gap for ViT-B/16 shows encoder-dependent scale bias despite stable ordering.

M.9 E4: Cross-Probe Consistency

Design. We compute the 6×6 pairwise SA_k alignment matrix M independently using ResNet-50 and ViT-B/16 on 4 common datasets {CIFAR-10, MNIST, STL-10, SVHN}. The 6 upper-triangle entries of each matrix are compared via Pearson r .

Results. Cross-probe Pearson $r = 0.891$ ($n = 6$, $p = 0.017$). Both probes agree on the direction of domain distance for all 6 pairs. CIFAR-10/STL-10 are close in both; CIFAR-10/MNIST are far in both. This small- n agreement does not separate dataset geometry from shared encoder biases and is reported only as a consistency check.

M.10 E_downstream: Effective Rank as Downstream Quality Proxy

Motivation. The Collapse Model predicts r_{eff} of merged features, but does higher r_{eff} actually correspond to better downstream performance? We test this by computing r_{eff} and linear-probe (LP) accuracy for 10 pretrained encoders on 5 datasets (50 total pairs).

Design. Encoders: ResNet-{50, 101}, EfficientNet-B4, ConvNeXt-T, DenseNet-121, ViT-B/16, DeiT-B, Swin-T, BEiT-B, RegNet-Y (covering CNN, ViT, and hybrid families). Datasets: CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST. For each (encoder, dataset) pair: extract $N=2000$ train features, compute r_{eff} ; train LP (sklearn LogisticRegression) on train, evaluate on test. Report per-dataset Spearman $\rho(r_{\text{eff}}, \text{LP acc})$ across 10 encoders.

Results.

Dataset	All 10 probes		ViT family ($d=768$)		CNN family ($d \geq 1024$)	
	n	ρ	n	ρ	n	ρ
CIFAR-10	10	+0.64	5	+0.90	5	+0.30
CIFAR-100	10	+0.61	5	+0.90	5	+0.30
STL-10	9	+0.73	5	+0.90	4	+0.20
SVHN	10	-0.03	5	+0.10	5	-0.30
Fashion-MNIST	10	+0.03	5	+0.60	5	-0.10

The “all probes” column ($\rho \approx +0.59$ on natural images) is confounded by feature dimension: CNNs ($d=2048$) have systematically higher r_{eff} than ViTs ($d=768$), inflating the apparent correlation. Controlling for dimension reveals a sharper picture: within the ViT family ($d=768$, 5 probes), r_{eff} is strongly predictive of LP accuracy on natural images ($\rho=0.90$ on CIFAR-10, CIFAR-100, and STL-10), while within the CNN family ($d \geq 1024$, 5 probes) the correlation is weak ($\rho=0.20-0.30$). This suggests that r_{eff} captures useful discriminative information primarily for architectures with compact feature spaces ($d < 1024$), where spectral diversity is more tightly coupled to representation quality.

Implication for the Collapse Model. r_{eff} should be understood as a measure of representational diversity—the number of independent directions activated by the data—rather than a direct proxy for downstream accuracy. Collapse-4S predicts when merging may increase this spectral diversity. Within one evaluated architecture family, diversity and accuracy

873 are positively correlated ($\rho=0.90$ for ViTs), but this association neither establishes necessity
874 nor guarantees that r_{eff} -based selection improves a new task. Across architecture families
875 or across datasets of varying difficulty, additional task-specific metrics are needed.

876 Sample-size sensitivity. Empirically, the over-prediction ratio grows with N/r_{eff} on 10 rep-
877 resentative ResNet-50 pairs: at $N=200$ the mean ratio is 1.021, rising monotonically to
878 1.545 at $N=2000$ (Spearman $\rho=0.80$ between N/r_{eff} and log bias). This association helps
879 predict calibration variation across encoders (ResNet-50 $c=0.78$, DINO $c=0.58$), but is not
880 asserted as a theoretical error law.

881 NeurIPS Paper Checklist

882 1. Claims

883 Question: Do the main claims made in the abstract and introduction accurately
884 reflect the paper’s contributions and scope?

885 Answer: [\[Yes\]](#)

886 Justification: The abstract and introduction (Section 1) separate exact Gram algebra and non-identifiability, a rank- L PSD sketch with deterministic error intervals, and an audited two-stage protocol. The abstract reports 100% selected-prefix interval coverage but zero certified decisions at $L \leq 50$, a competitive DPP baseline, and the source-supervised adapter’s significant loss to frozen identity. It explicitly disclaims global optimality, privacy, practical compression, and downstream benefit.

893 Guidelines:

- 894 • The answer [\[N/A\]](#) means that the abstract and introduction do not include the
895 claims made in the paper.
- 896 • The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- 900 • The claims made should match theoretical and experimental results, and reflect
901 how much the results can be expected to generalize to other settings.
- 902 • It is fine to include aspirational goals as motivation as long as it is clear that
903 these goals are not attained by the paper.

904 2. Limitations

905 Question: Does the paper discuss the limitations of the work performed by the
906 authors?

907 Answer: [\[Yes\]](#)

908 Justification: Appendix A states that marginal spectra and principal angles are insufficient in general, Collapse-4S has no universal guarantee, and the rank- L interval certifies none of the tested choices. It also discloses unequal communication budgets, numerical conditioning, encoder-dependent calibration, weak cross-modal transfer, limited domain balance, failure of the source-supervised adapter against identity, and the absence of global-optimality, privacy, fairness, or downstream-utility guarantees.

915 Guidelines:

- 916 • The answer [\[N/A\]](#) means that the paper has no limitation while the answer
917 [\[No\]](#) means that the paper has limitations, but those are not discussed in the
918 paper.
- 919 • The authors are encouraged to create a separate “Limitations” section in their
920 paper.
- 921 • The paper should point out any strong assumptions and how robust the results
922 are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- 926 • The authors should reflect on the scope of the claims made, e.g., if the approach
927 was only tested on a few datasets or with a few runs. In general, empirical
928 results often depend on implicit assumptions, which should be articulated.
- 929 • The authors should reflect on the factors that influence the performance of
930 the approach. For example, a facial recognition algorithm may perform poorly
931 when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[Yes\]](#)

Justification: Proposition 1, Lemma 2, and Proposition 3 distinguish the unconditional Gram identity from the block-compatible and proportional-spectrum special cases. Proposition 5 gives an explicit counterexample to identification even from complete marginal spectra and principal angles. Theorem 1 and Corollary 6 state the rank- L tail assumptions, deterministic error interval, and sufficient one-step certificate. Equation 8 is labelled an empirical approximation. Proofs and optimization limits are provided in Appendices B and C.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: Algorithm 1 gives the common greedy loop, while Section 5 specifies the corrected Stage-1 and Stage-2 information boundaries, equal-source weighting, baselines, controls, split discipline, and reported metrics. Appendix M documents the historical 427-pair study and its hyperparameters. The released corrected scripts record feature, index, model-state, preprocessing, code, and configuration hashes and reject incomplete or provenance-inconsistent result shards.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- If the paper includes experiments, a [\[No\]](#) answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code and instructions are provided at the anonymous repository <https://anonymous.4open.science/r/collapsemodel-1DB6>. The historical experiments use `cd code && python run_all.py` with `code/configs/repro_config.yaml`. The root and corrected-result READMEs list the feature-extraction, controlled rank-sweep, screening, source-supervised validation, held-out, and aggregation commands. Raw benchmarks and pre-trained weights are obtained from their original public distributions and are not redistributed.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.

- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Section 4 and Appendix M specify the historical study. Section 5 separately specifies label-blind source sampling, source-supervised adapter training, target-train-only cross-validation, frozen held-out evaluation, seeds, weighting, shortlist budgets, baselines, and identity/random-adapter controls. It also states that every adapter was trained offline to estimate exhaustive regret, so no realized deployment saving is claimed. Exact settings are stored in each run manifest.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: Bootstrap confidence intervals ($B=10,000$) are reported for the historical correlation statistics, including the non-significant $\Delta\rho=0.003$ gap over r_A+r_B . The corrected Stage-2 control uses target-cluster bootstrap; oracle minus frozen identity is -0.0670 with 95% CI $[-0.0977, -0.0440]$. The rank- L coverage statement is a deterministic check over all evaluated prefixes, and no inferential superiority claim is made from the controlled gain table.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The authors should answer [\[Yes\]](#) if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.

1095 • For asymmetric distributions, the authors should be careful not to show in
1096 tables or figures symmetric error bars that would yield results that are out of
1097 range (e.g., negative error rates).
1098 • If error bars are reported in tables or plots, the authors should explain in the
1099 text how they were calculated and reference the corresponding figures or tables
1100 in the text.

1101 8. Experiments compute resources

1102 Question: For each experiment, does the paper provide sufficient information on the
1103 computer resources (type of compute workers, memory, time of execution) needed
1104 to reproduce the experiments?

1105 Answer: [Yes]

1106 Justification: The historical reproduction config
1107 (code/configs/repro_config.yaml) reports per-experiment estimates and
1108 an approximately eight-hour single-CUDA-node total. The corrected two-stage
1109 README and manifests separately record CPU selection time, Ascend 910B4
1110 feature/adaptation requirements, device assignment, run counts, and observed
1111 wall-clock time. These estimates are hardware-specific; ViT-B/16 feature extrac-
1112 tion on the reported Ascend system includes CPU fallback and is labelled as mixed
1113 CPU/NPU wall time.

1114 Guidelines:

1115 • The answer [N/A] means that the paper does not include experiments.
1116 • The paper should indicate the type of compute workers CPU or GPU, internal
1117 cluster, or cloud provider, including relevant memory and storage.
1118 • The paper should provide the amount of compute required for each of the
1119 individual experimental runs as well as estimate the total compute.
1120 • The paper should disclose whether the full research project required more com-
1121 pute than the experiments reported in the paper (e.g., preliminary or failed
1122 experiments that didn't make it into the paper).

1123 9. Code of ethics

1124 Question: Does the research conducted in the paper conform, in every respect, with
1125 the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

1126 Answer: [Yes]

1127 Justification: The research analyzes frozen representations and evaluates dataset-
1128 pool screening on existing benchmarks and pretrained encoders. It involves no newly
1129 recruited human subjects and releases no generated content or new personal-data
1130 corpus. Potential selection bias, privacy overstatement, and inappropriate use of a
1131 spectral proxy as a utility decision are disclosed as limitations.

1132 Guidelines:

1133 • The answer [N/A] means that the authors have not reviewed the NeurIPS Code
1134 of Ethics.
1135 • If the authors answer [No], they should explain the special circumstances that
1136 require a deviation from the Code of Ethics.
1137 • The authors should make sure to preserve anonymity (e.g., if there is a special
1138 consideration due to laws or regulations in their jurisdiction).

1139 10. Broader impacts

1140 Question: Does the paper discuss both potential positive societal impacts and neg-
1141 ative societal impacts of the work performed?

1142 Answer: [Yes]

1143 Justification: The method can reduce communication by transmitting sketches in-
1144 stead of sample rows, but this is not a differential-privacy guarantee and sketches
1145 may still leak information. Spectral diversity may improve pool coverage, yet it can
1146 also preserve, hide, or amplify biases in the encoder and candidate datasets and
1147 may underselect small groups. We therefore require task-aware validation and do
1148 not claim that the criterion is content-neutral or fairness-preserving.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No high-risk generative model, scraped corpus, or pretrained checkpoint is released. The code computes spectral summaries and trains only small evaluation adapters on user-supplied features. Ordinary access control and dataset/model terms remain the user’s responsibility; the paper does not claim that the summaries are privacy-preserving.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The paper cites the original publications and distributions for the benchmark datasets and pretrained encoder families used in each experiment. The

repository does not redistribute those assets; reproduction requires users to obtain them from the original providers and comply with the provider-specific licenses and terms. We do not make the unsupported claim that every benchmark has the same or a uniformly permissive license.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The paper introduces no new dataset or pretrained model. The new reproducibility code is documented in the repository README, command-level reproduction guide, experiment plan, audit log, and machine-readable manifests; its scope and current empirical limitations are stated explicitly.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects. All experiments use existing public benchmark datasets with automated evaluation.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

1255 • According to the NeurIPS Code of Ethics, workers involved in data collection,
1256 curation, or other labor should be paid at least the minimum wage in the
1257 country of the data collector.

1258 15. Institutional review board (IRB) approvals or equivalent for research with human
1259 subjects

1260 Question: Does the paper describe potential risks incurred by study participants,
1261 whether such risks were disclosed to the subjects, and whether Institutional Review
1262 Board (IRB) approvals (or an equivalent approval/review based on the requirements
1263 of your country or institution) were obtained?

1264 Answer: [N/A]

1265 Justification: The paper does not involve human subjects research. All data used
1266 are existing public benchmarks.

1267 Guidelines:

1268 • The answer [N/A] means that the paper does not involve crowdsourcing nor
1269 research with human subjects.

1270 • Depending on the country in which research is conducted, IRB approval (or
1271 equivalent) may be required for any human subjects research. If you obtained
1272 IRB approval, you should clearly state this in the paper.

1273 • We recognize that the procedures for this may vary significantly between insti-
1274 tutions and locations, and we expect authors to adhere to the NeurIPS Code
1275 of Ethics and the guidelines for their institution.

1276 • For initial submissions, do not include any information that would break
1277 anonymity (if applicable), such as the institution conducting the review.

1278 16. Declaration of LLM usage

1279 Question: Does the paper describe the usage of LLMs if it is an important, original,
1280 or non-standard component of the core methods in this research? Note that if the
1281 LLM is used only for writing, editing, or formatting purposes and does not impact
1282 the core methodology, scientific rigor, or originality of the research, declaration is
1283 not required.

1284 Answer: [N/A]

1285 Justification: The core methodology (PSD Gram sketches, spectral decomposition,
1286 and fixed-compute adapter evaluation) does not use a large language model. LLMs
1287 are not part of the proposed method or experimental pipeline.

1288 Guidelines:

1289 • The answer [N/A] means that the core method development in this research
1290 does not involve LLMs as any important, original, or non-standard components.

1291 • Please refer to our LLM policy in the NeurIPS handbook for what should or
1292 should not be described.